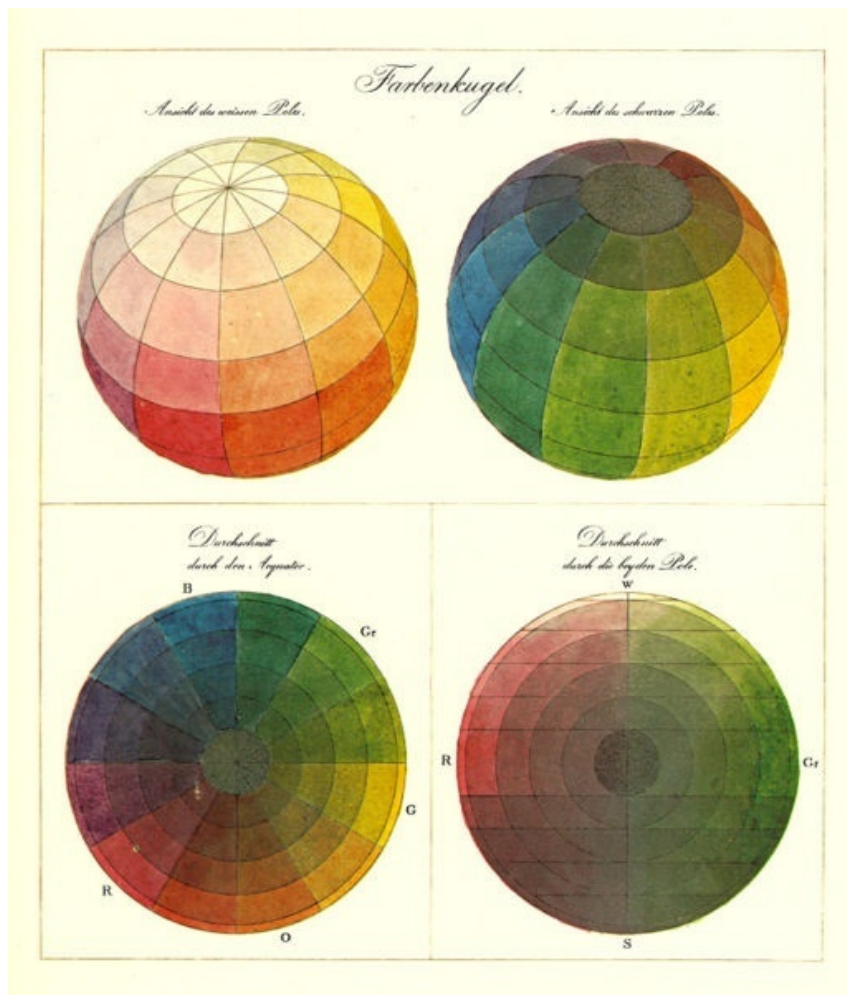


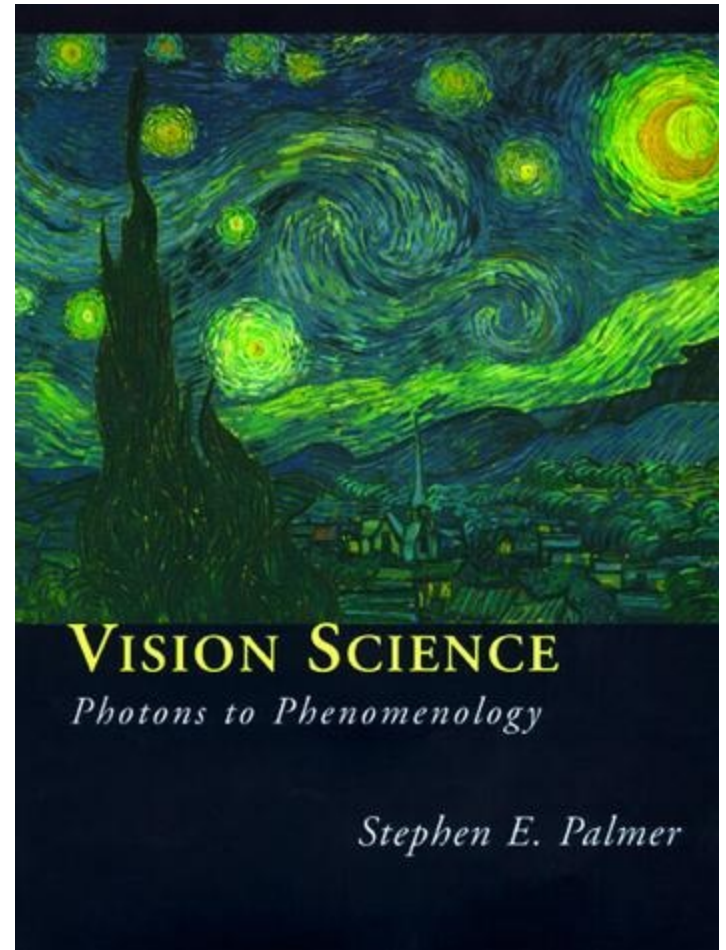
Color



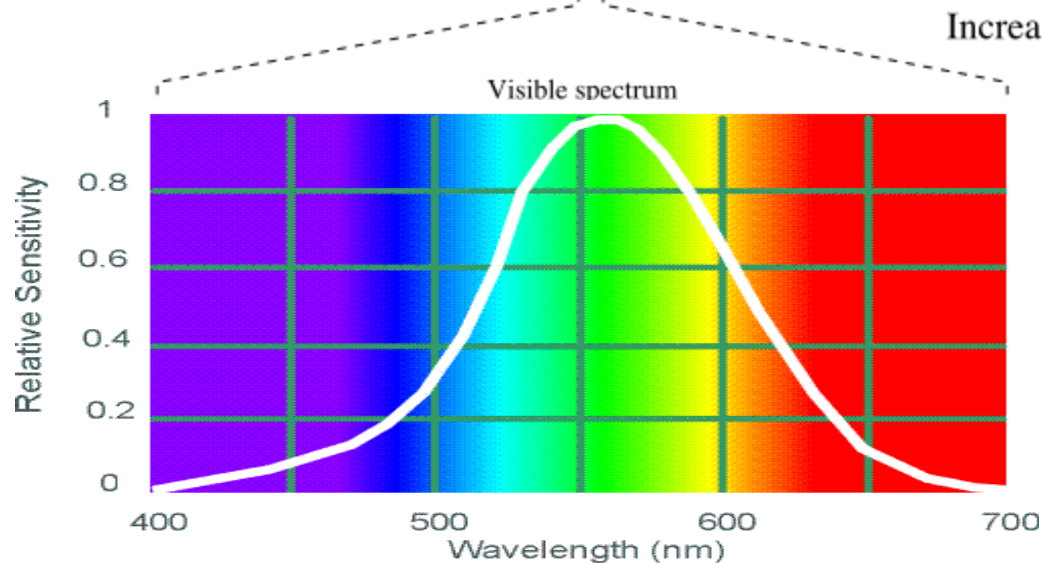
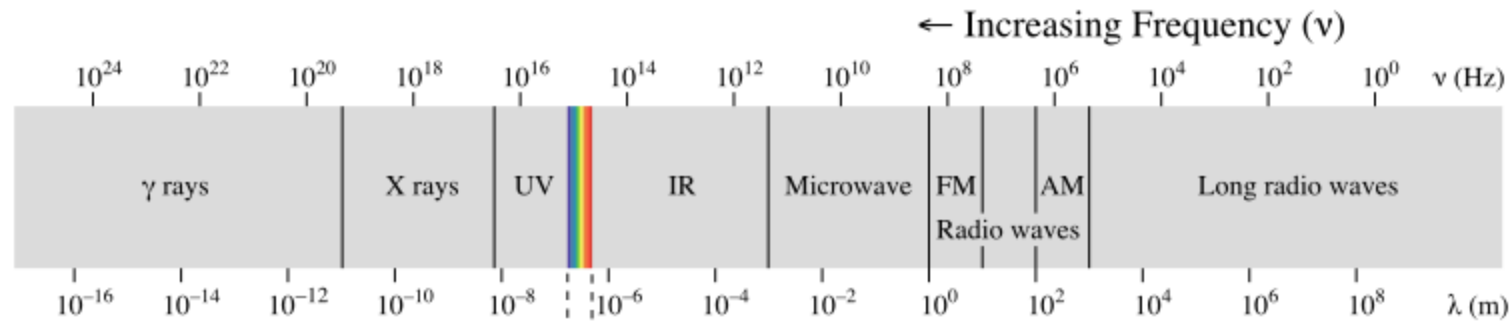
Phillip Otto Runge (1777-1810)

What is color?

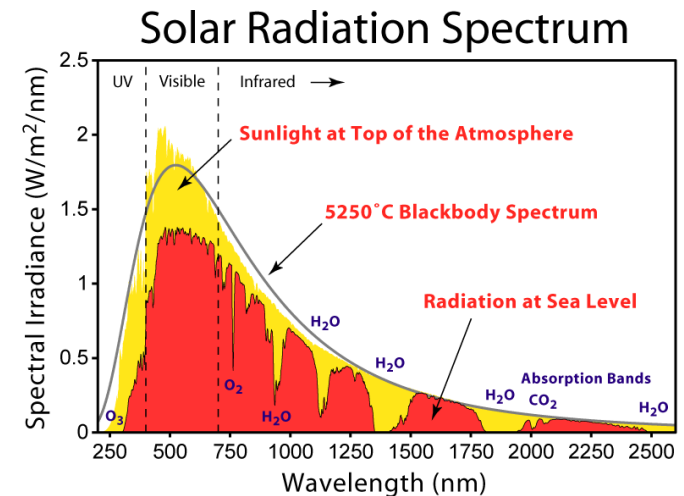
- Color is the result of interaction between physical light in the environment and our visual system
- Color is a psychological property of our visual experiences when we look at objects and lights, *not* a physical property of those objects or lights (S. Palmer, *Vision Science: Photons to Phenomenology*)



Electromagnetic spectrum

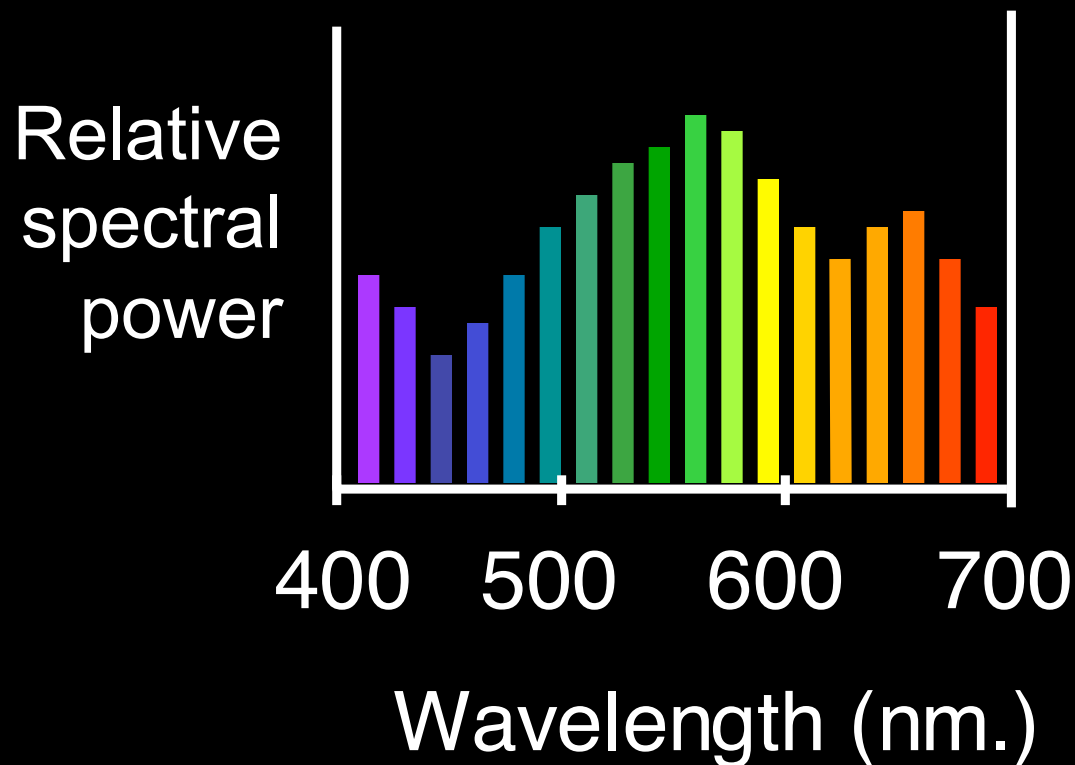


Human Luminance Sensitivity Function



The Physics of Light

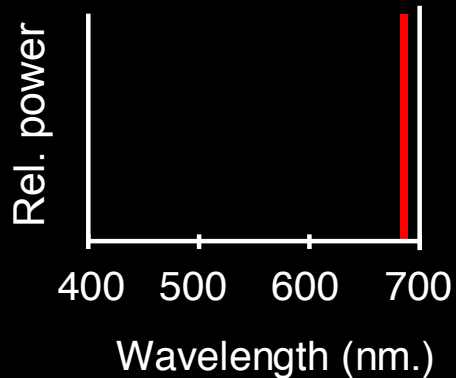
Any source of light can be completely described physically by its spectrum: the amount of energy emitted (per time unit) at each wavelength 400 - 700 nm.



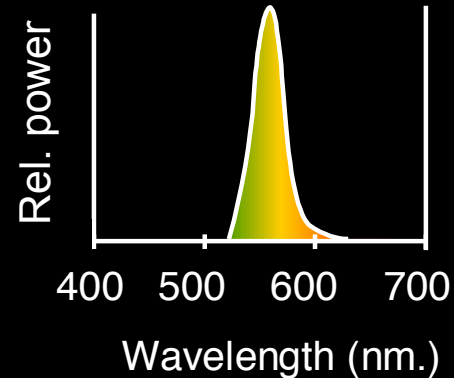
Spectra of Light Sources

Some examples of the spectra of light sources

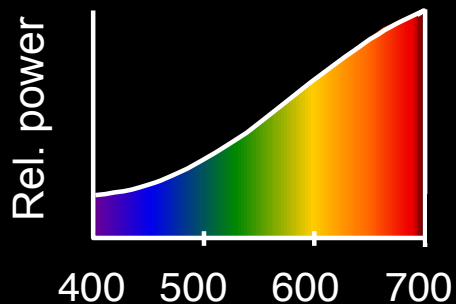
A. Ruby Laser



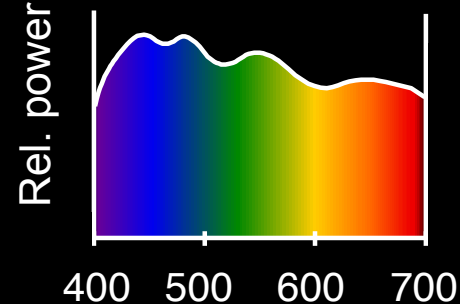
B. Gallium Phosphide Crystal



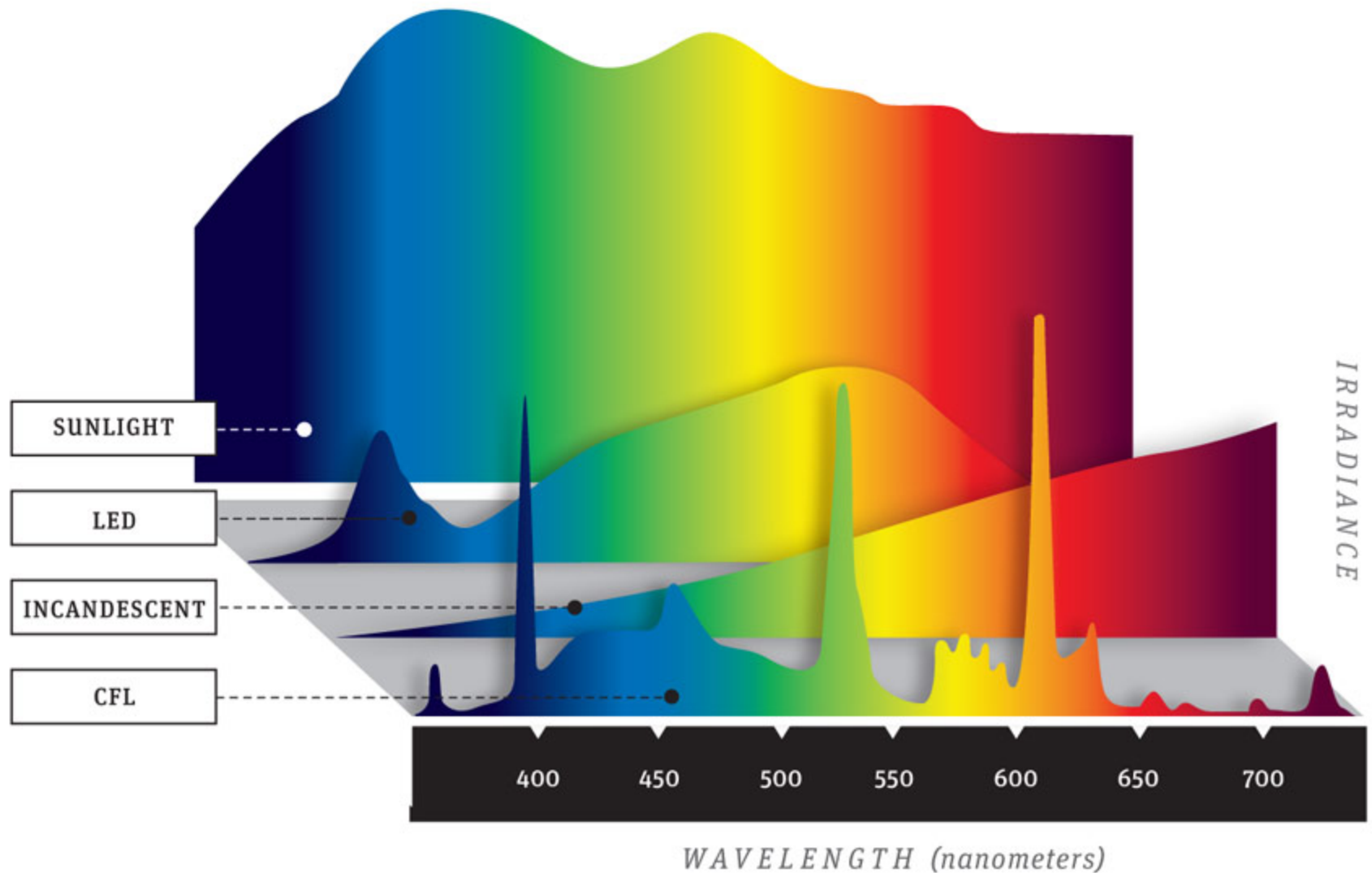
C. Tungsten Lightbulb



D. Normal Daylight

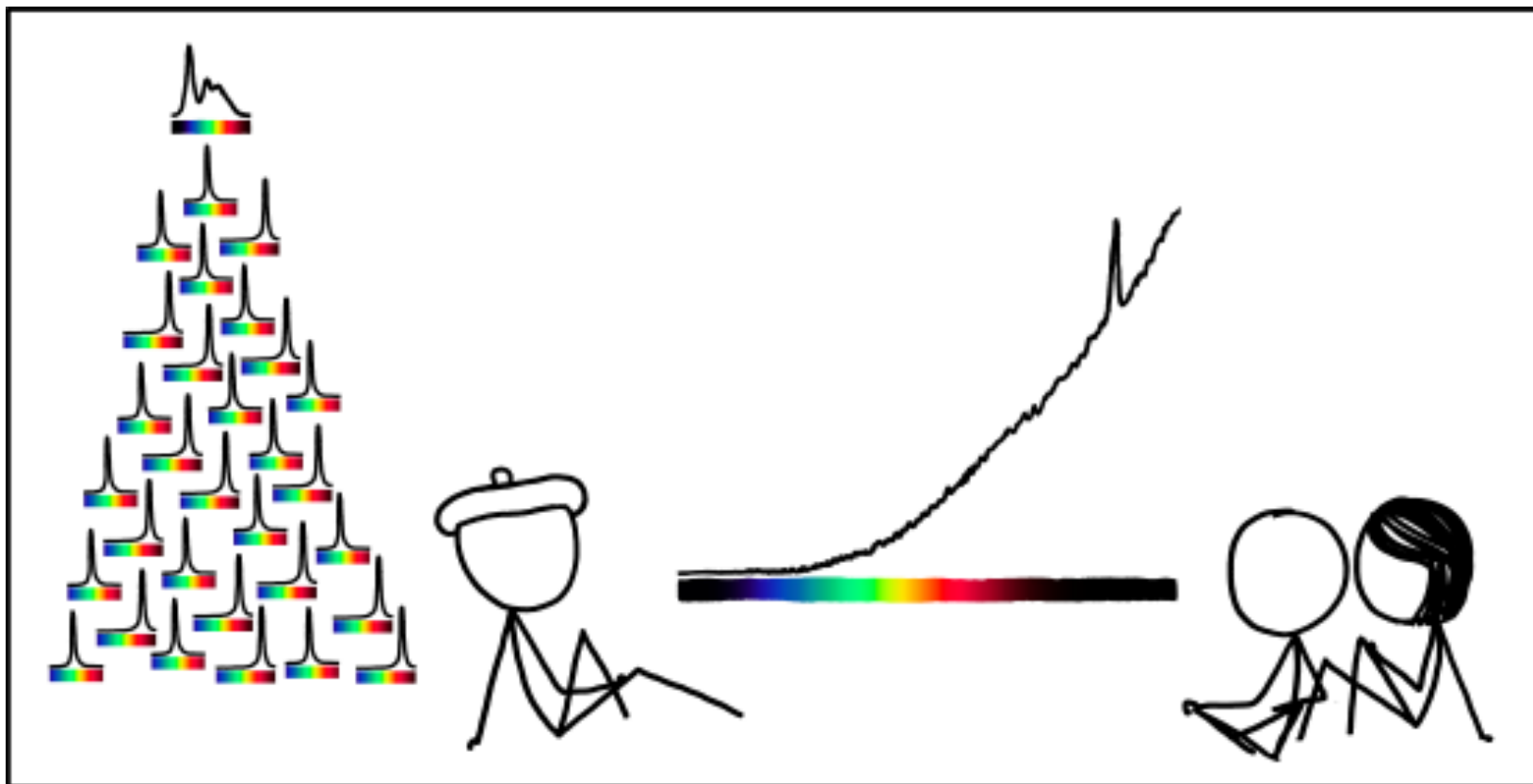


Spectra of light sources



[Source: Popular Mechanics](#)

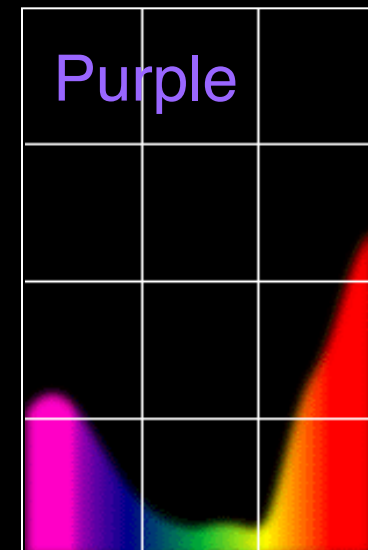
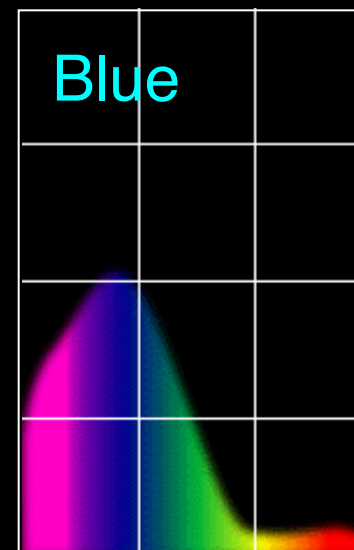
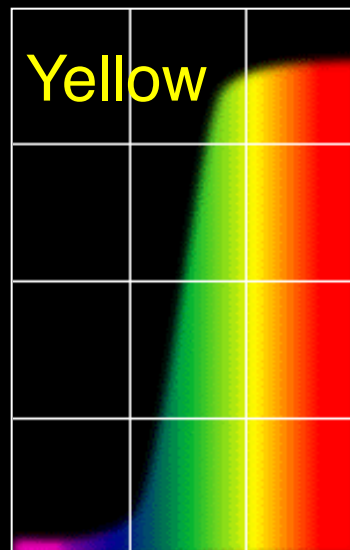
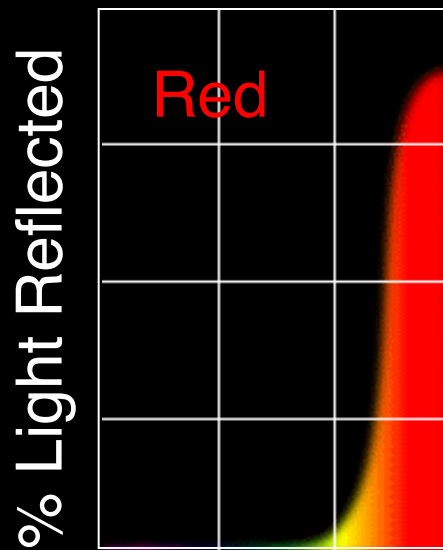
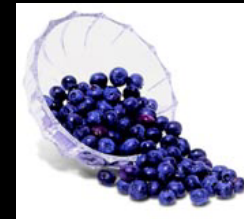
XKCD Christmas Lights



<http://www.xkcd.com/1308/>

Reflectance Spectra of Surfaces

Some examples of the reflectance spectra of surfaces



400

700

400

700

400

700

400

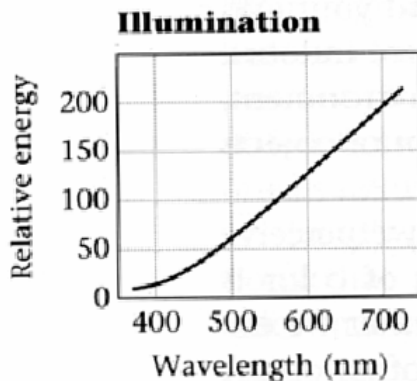
700

Wavelength (nm)

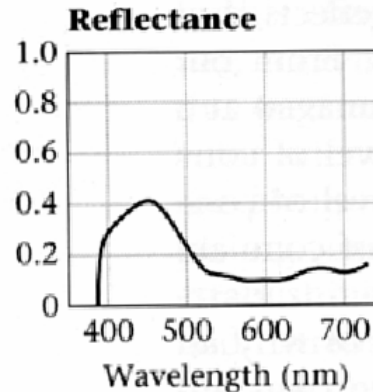
Interaction of light and surfaces



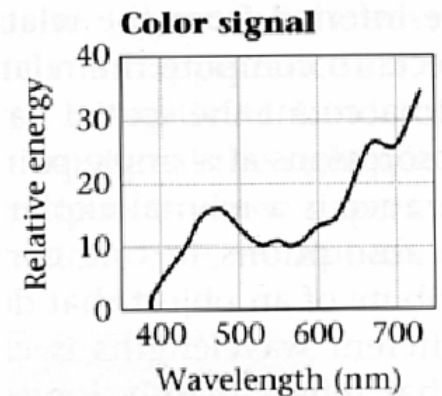
- Reflected color is the result of interaction of light source spectrum with surface reflectance



• *



=



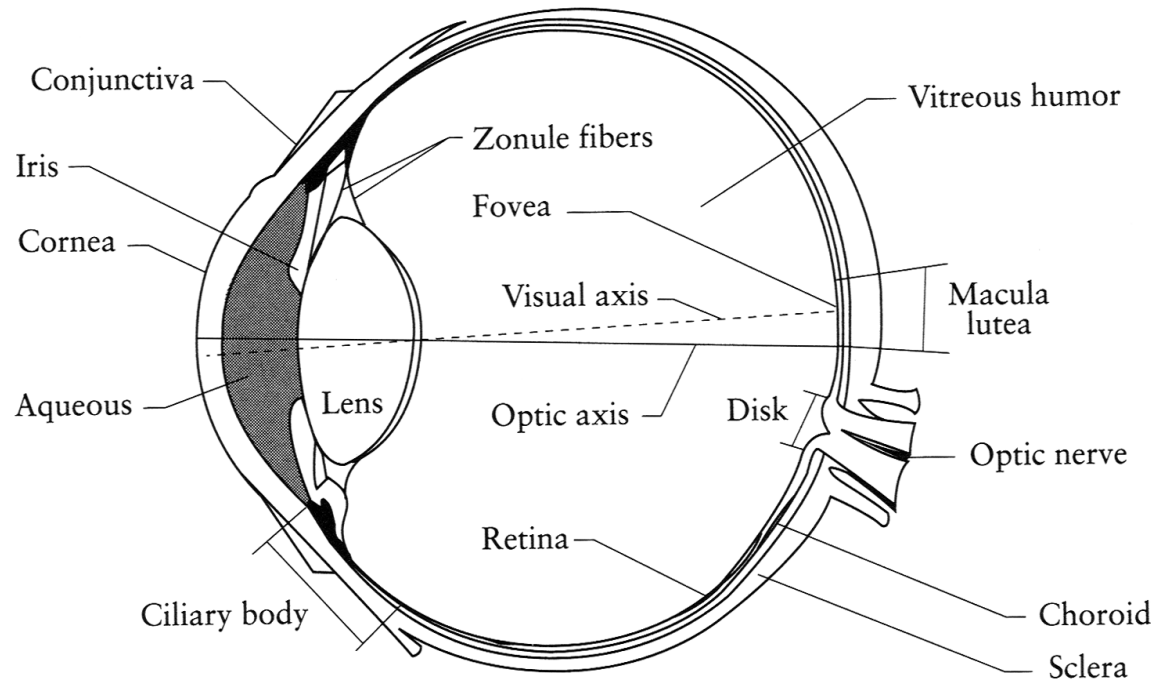
Interaction of light and surfaces

- What is the observed color of any surface under monochromatic light?



Olafur Eliasson, *Room for one color*

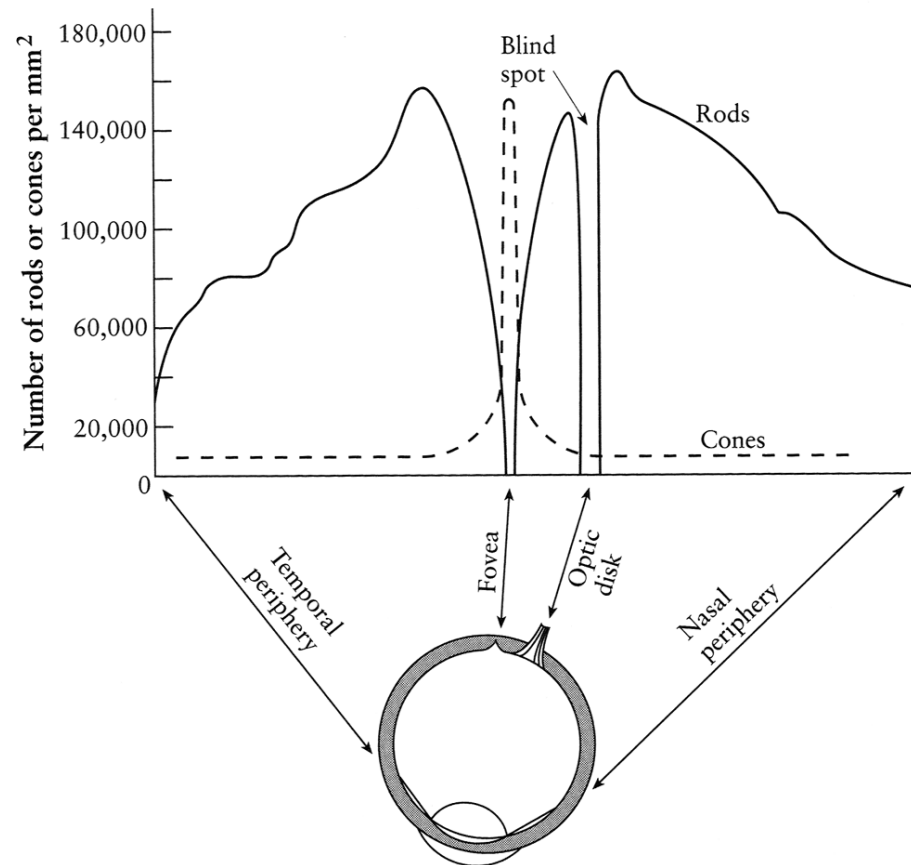
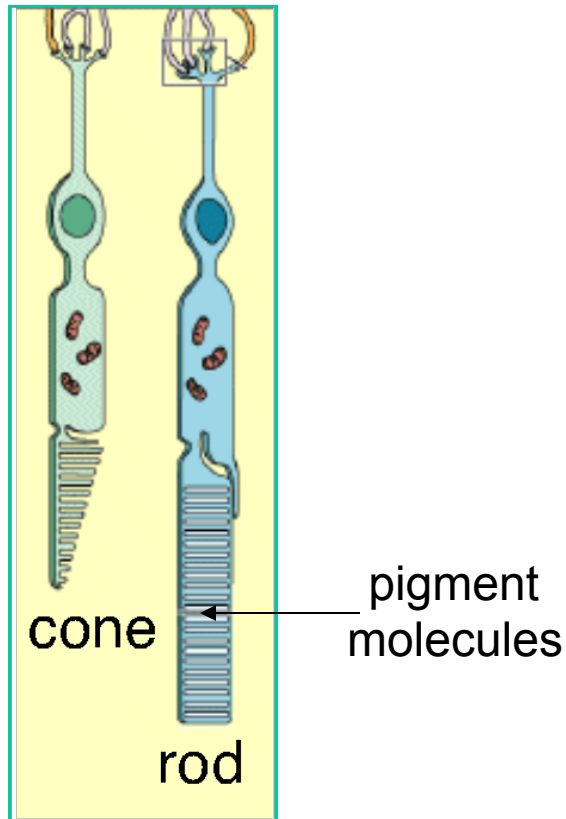
The Eye



The human eye is a camera!

- **Lens** - changes shape by using ciliary muscles (to focus on objects at different distances)
- **Pupil** - the hole (aperture) whose size is controlled by the iris
- **Iris** - colored annulus with radial muscles
- **Retina** - photoreceptor cells

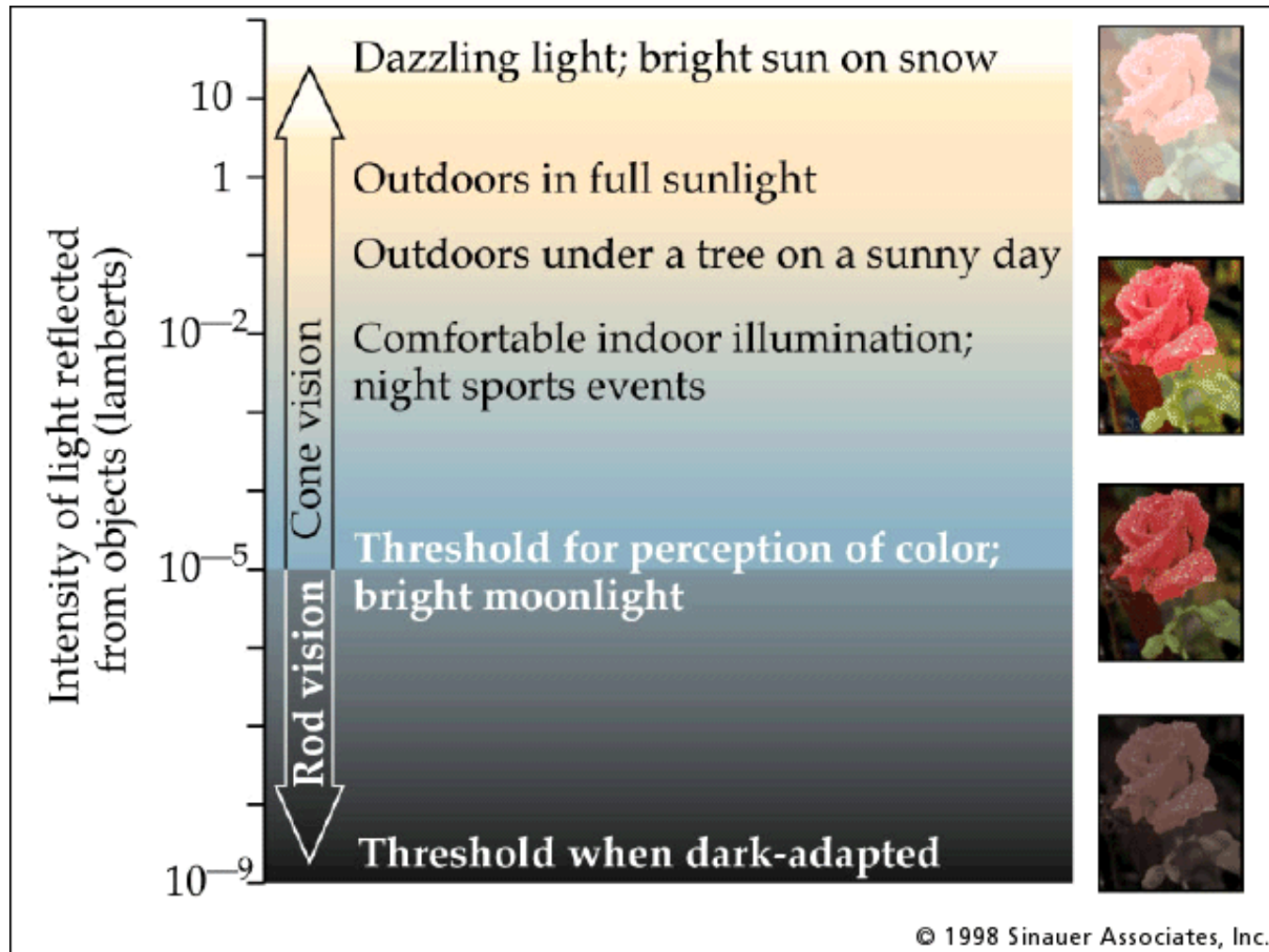
Rods and cones, fovea



Rods are responsible for intensity, cones for color perception
Rods and cones are *non-uniformly* distributed on the retina

- **Fovea** - Small region (1 or 2°) at the center of the visual field containing the highest density of cones – *and no rods*

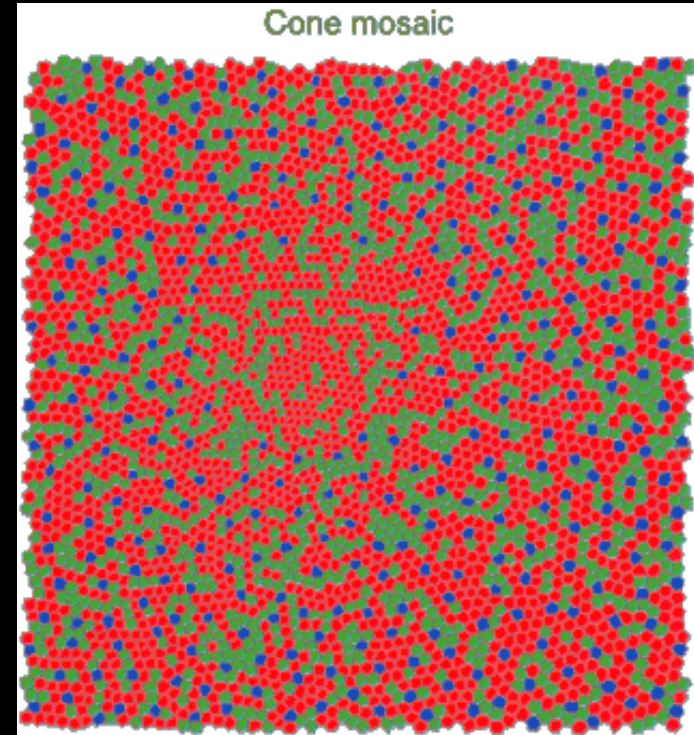
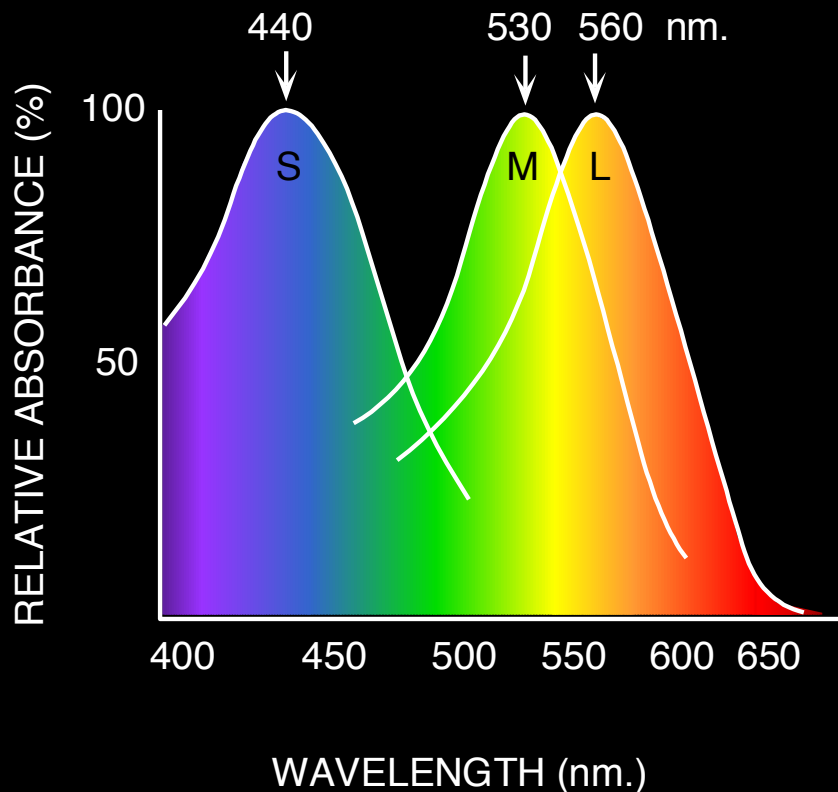
Rod / Cone sensitivity



Why can't we read in the dark?

Physiology of Color Vision

Three kinds of cones:



- Ratio of L to M to S cones: approx. 10:5:1
- Almost no S cones in the center of the fovea

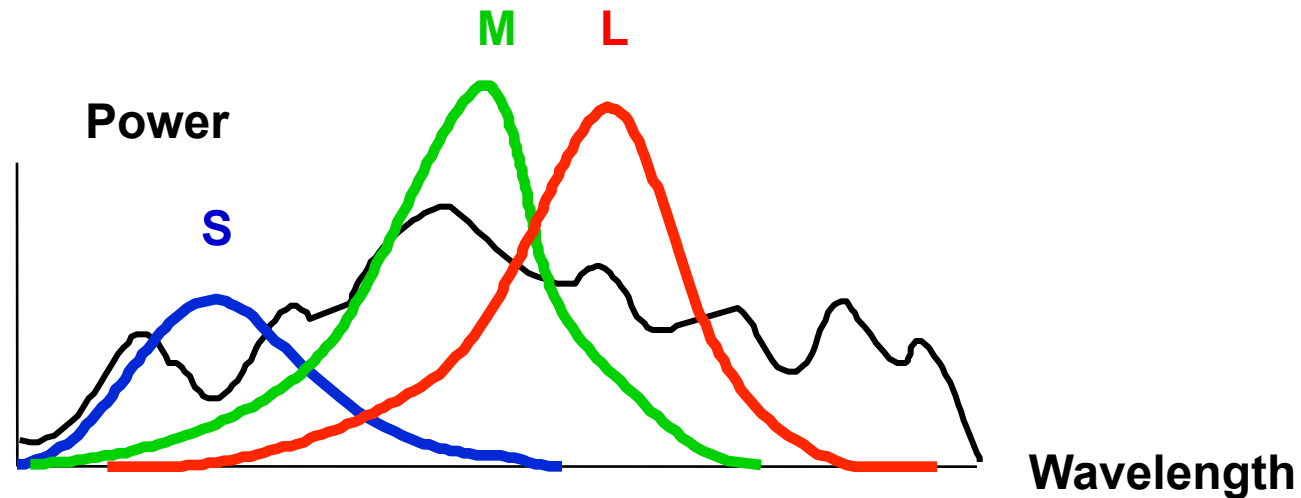
Physiology of Color Vision: Fun facts

- “M” and “L” pigments are encoded on the X-chromosome
 - That’s why men are more likely to be color blind
 - “L” gene has high variation, so some women may be *tetrachromatic*
- Some animals have one (night animals), two (e.g., dogs), four (fish, birds), five (pigeons, some reptiles/amphibians), or even 12 (mantis shrimp) types of cones

<http://ngm.nationalgeographic.com/2016/02/evolution-of-eyes-text>

http://en.wikipedia.org/wiki/Color_vision

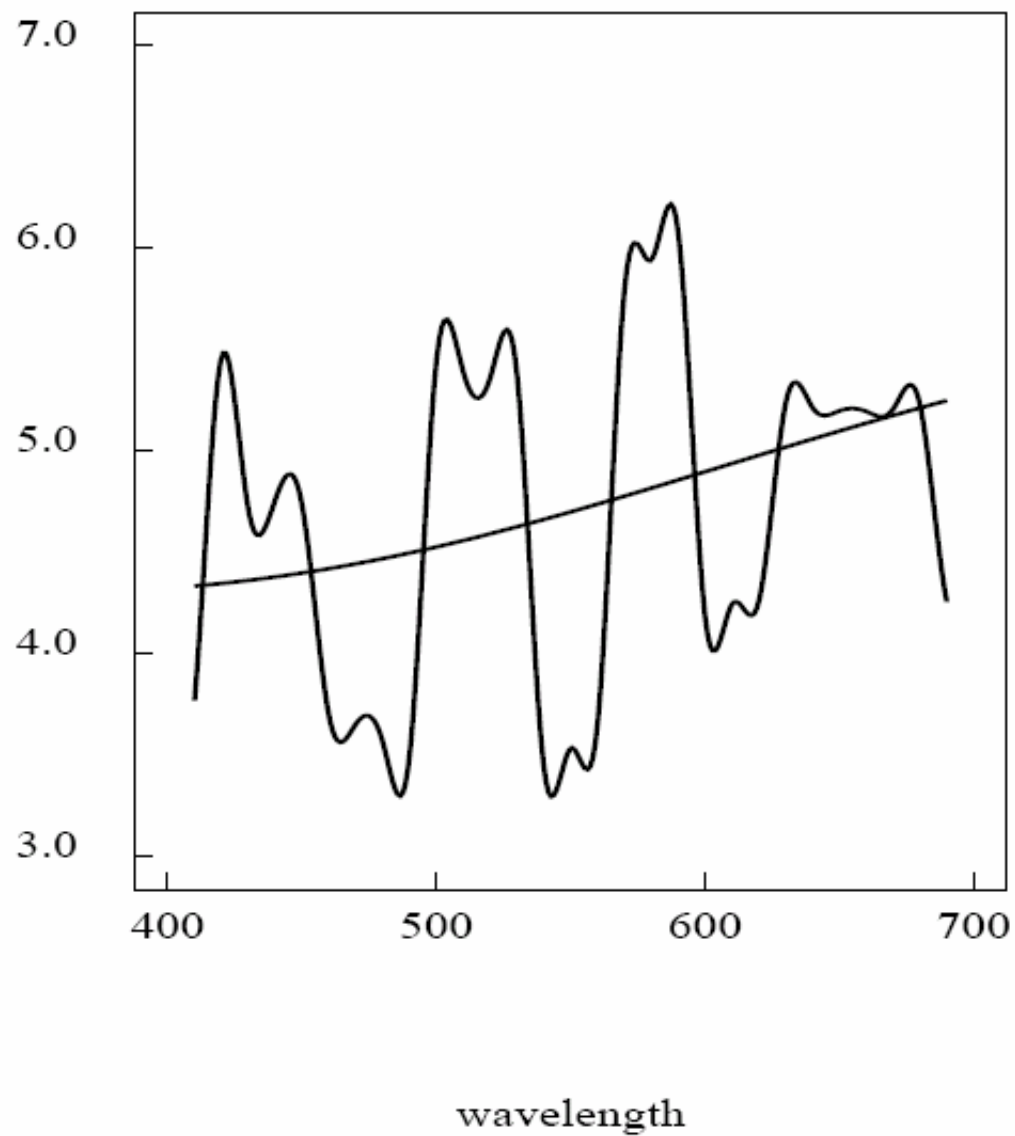
Color perception



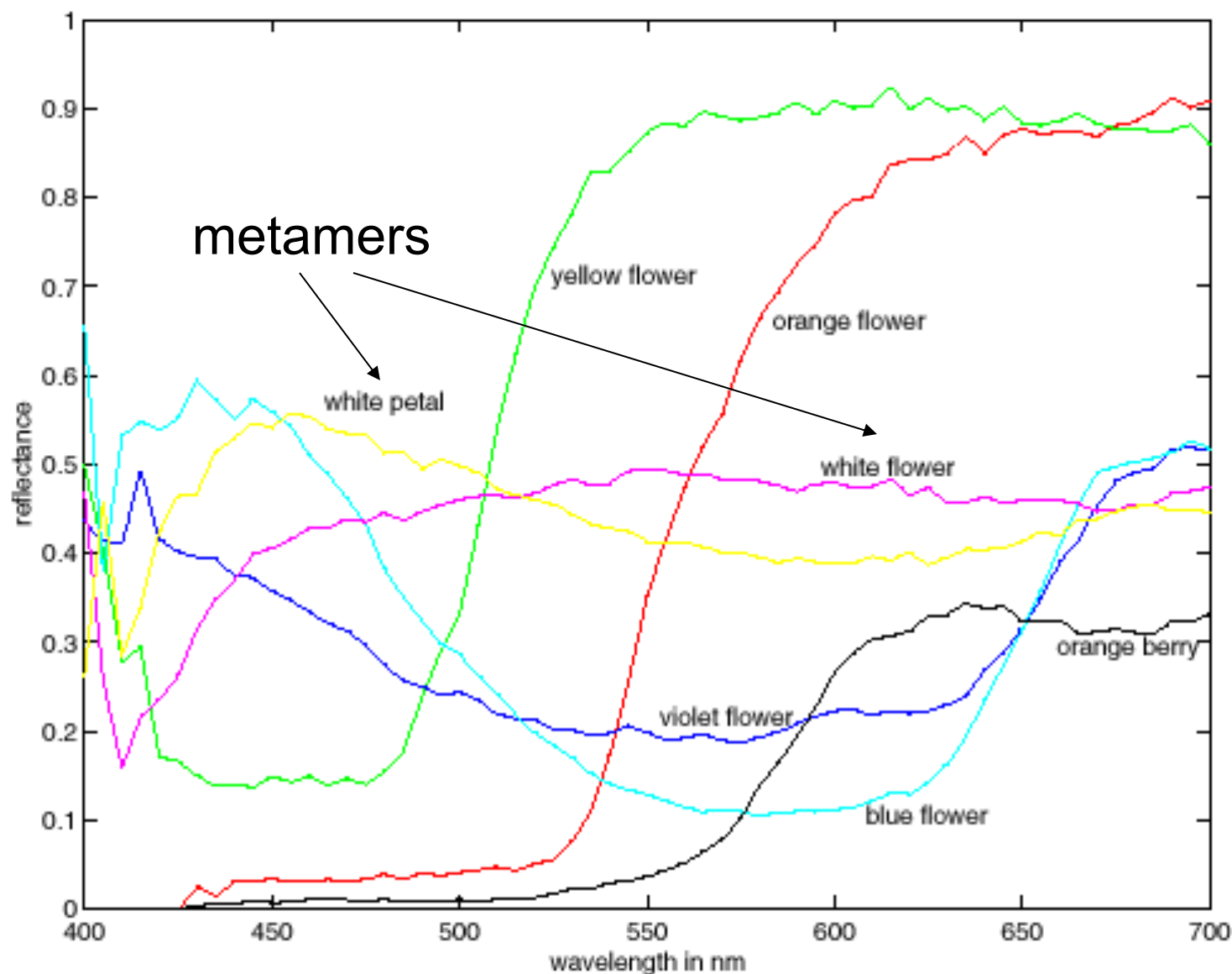
Rods and cones act as *filters* on the spectrum

- To get the output of a filter, multiply its response curve by the spectrum, integrate over all wavelengths
 - Each cone yields one number
- How can we represent an entire spectrum with 3 numbers?
- We can't! Most of the information is lost
 - As a result, two different spectra may appear indistinguishable
 - » such spectra are known as **metamers**

Metamers

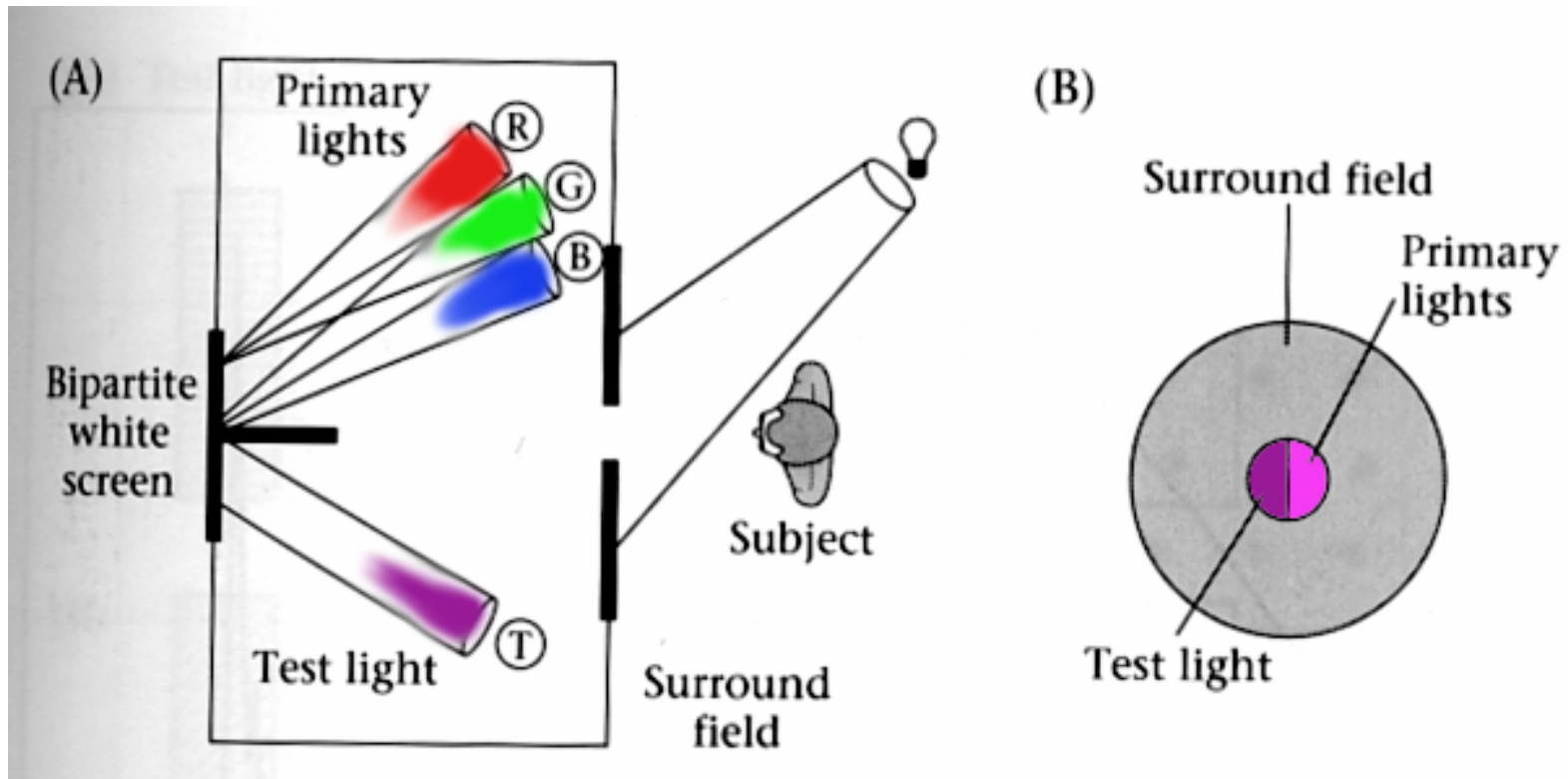


Spectra of some real-world surfaces

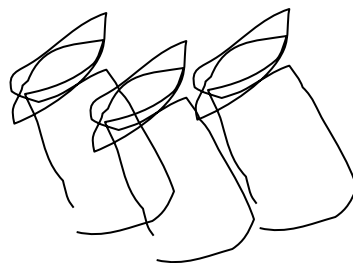
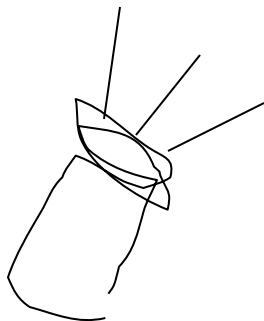
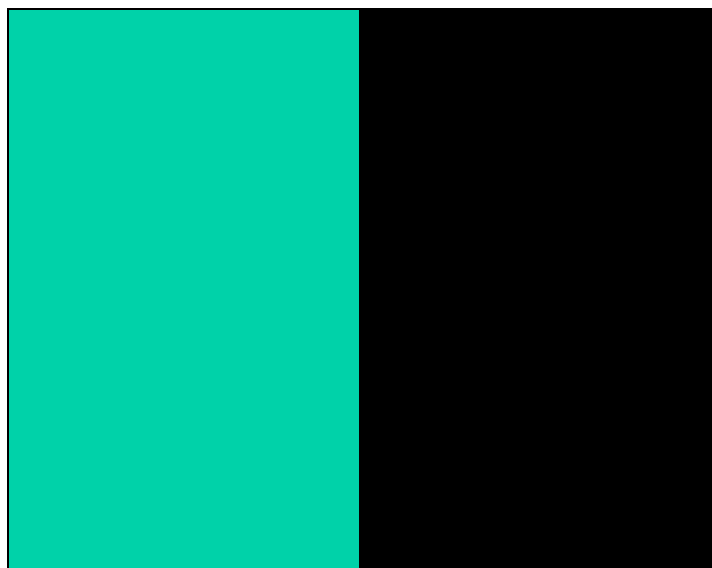


Standardizing color experience

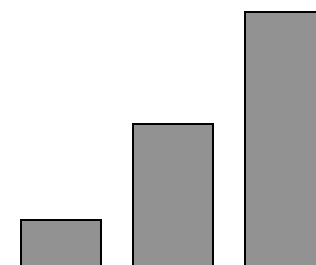
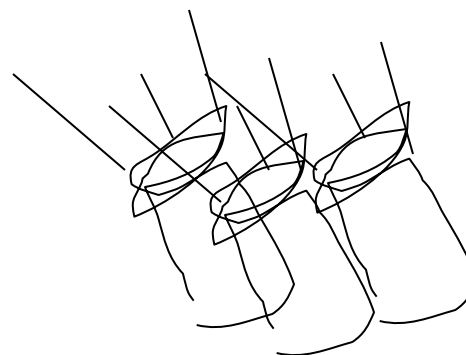
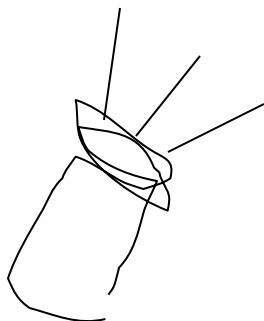
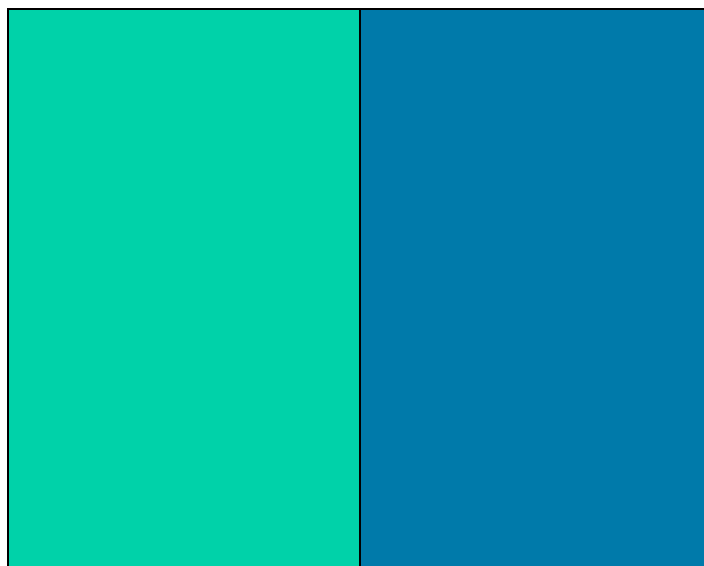
- We would like to understand which spectra produce the same color sensation in people under similar viewing conditions
- Color matching experiments



Color matching experiment 1



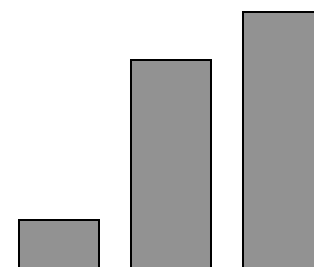
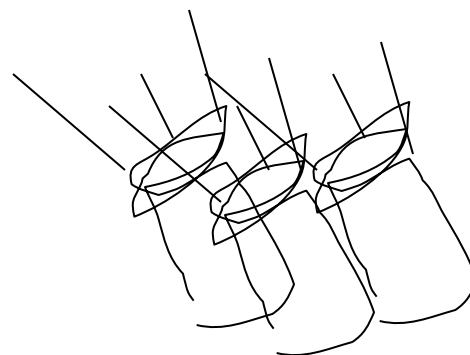
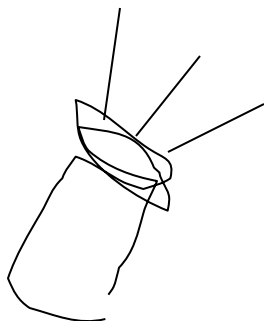
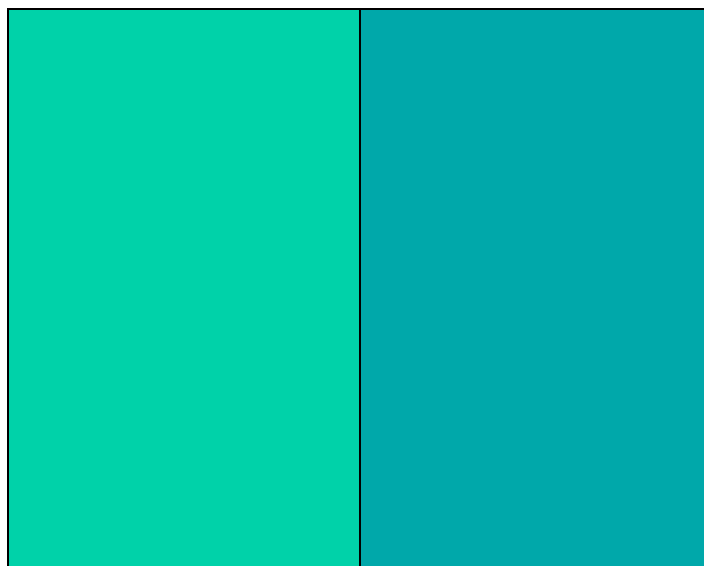
Color matching experiment 1



p_1 p_2 p_3

Source: W. Freeman

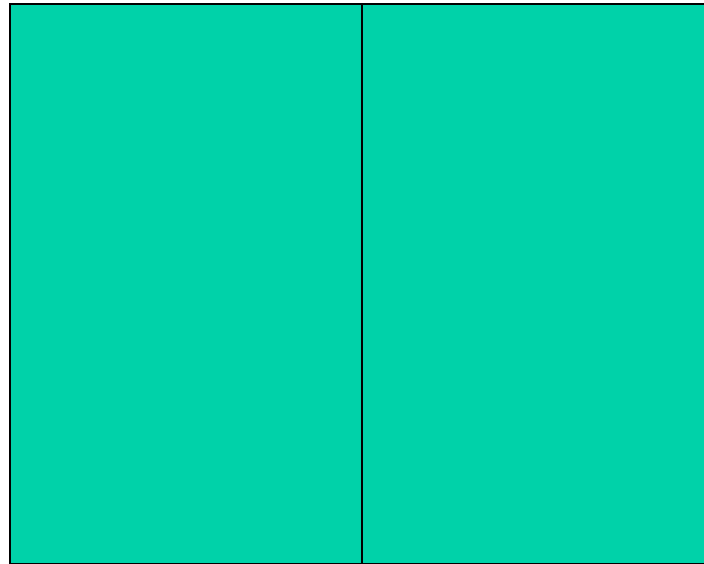
Color matching experiment 1



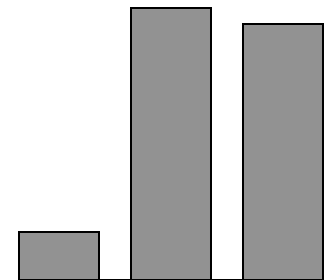
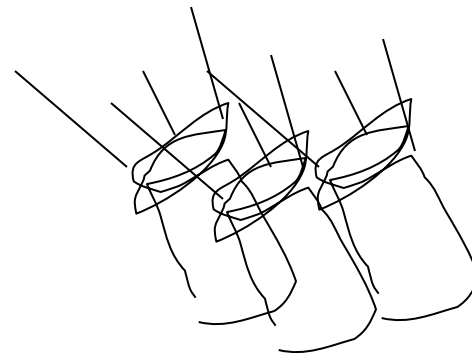
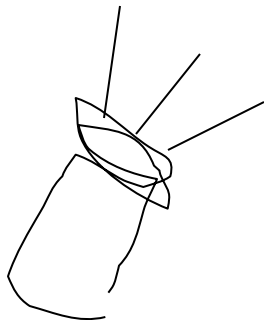
p_1 p_2 p_3

Source: W. Freeman

Color matching experiment 1



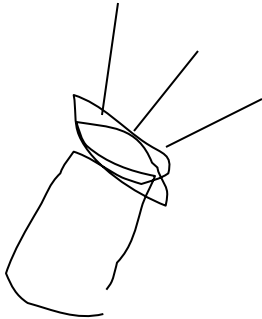
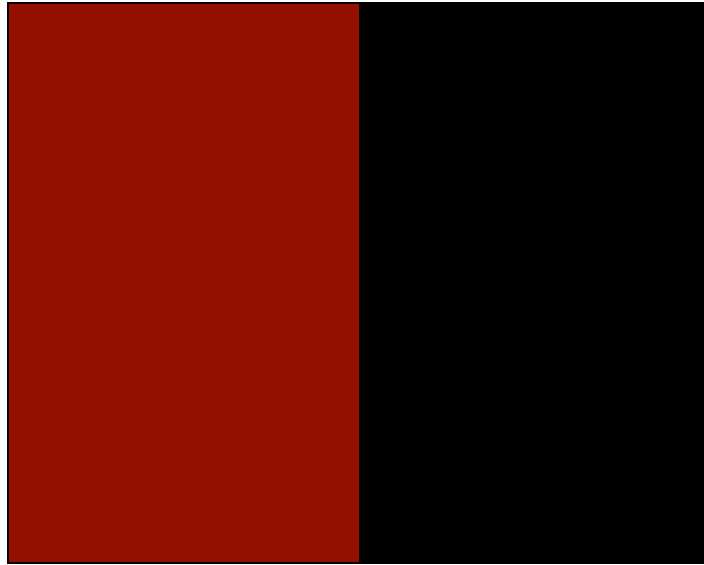
The primary color amounts needed for a match



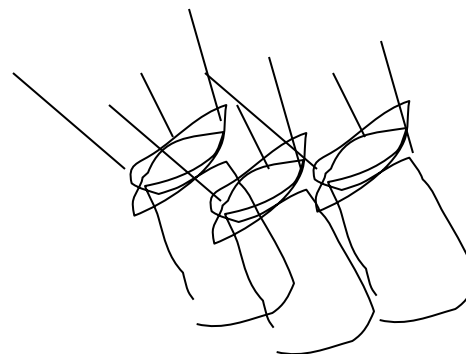
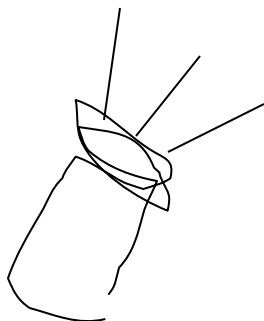
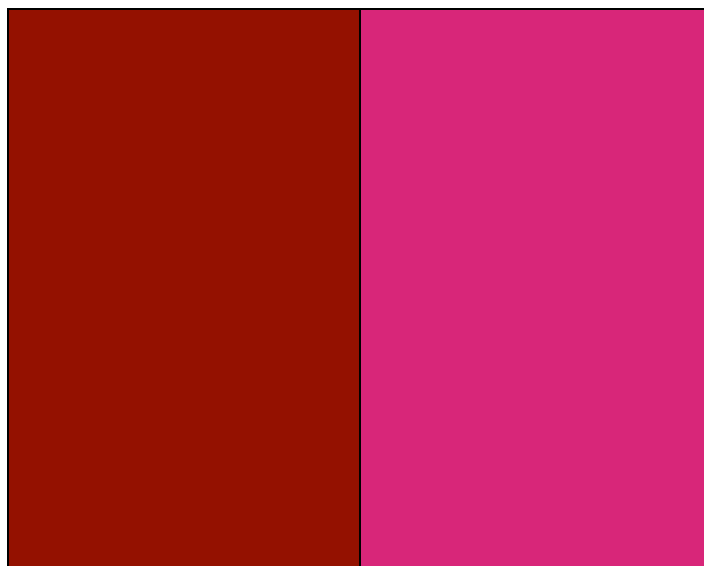
p_1 p_2 p_3

Source: W. Freeman

Color matching experiment 2



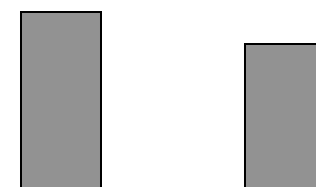
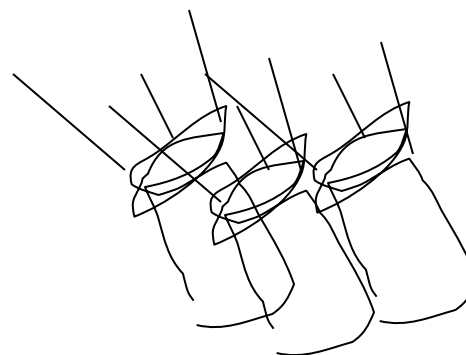
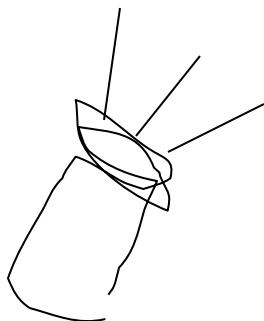
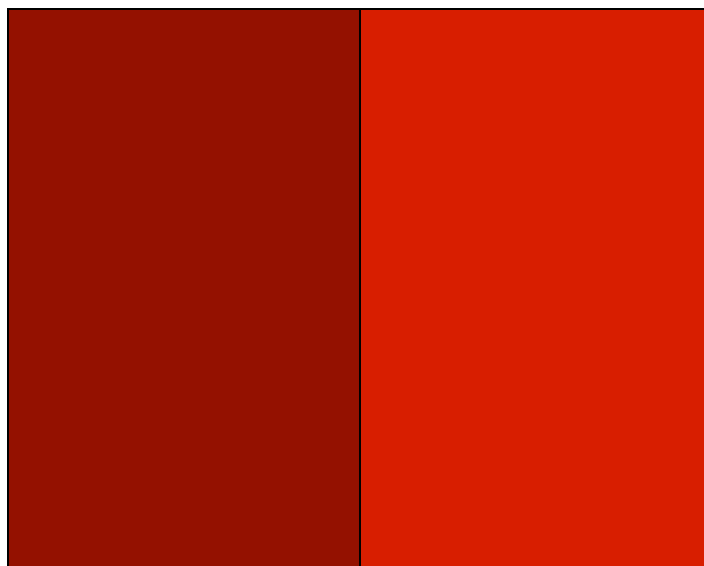
Color matching experiment 2



p_1 p_2 p_3

Source: W. Freeman

Color matching experiment 2

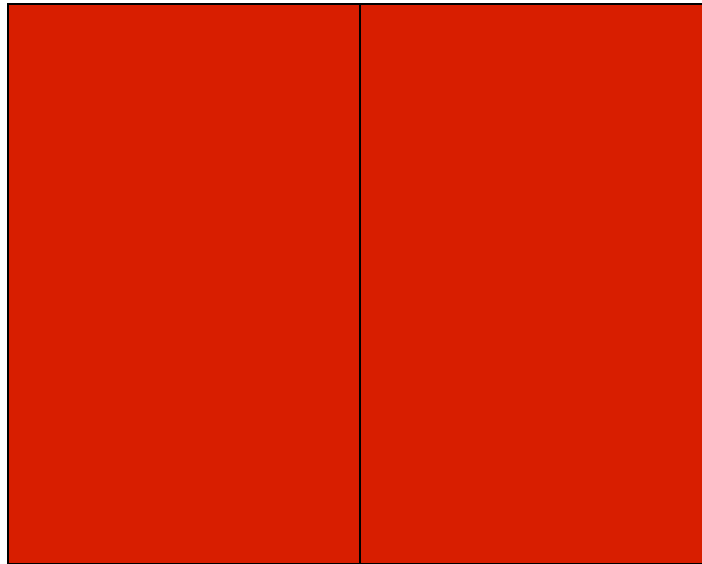


p_1 p_2 p_3

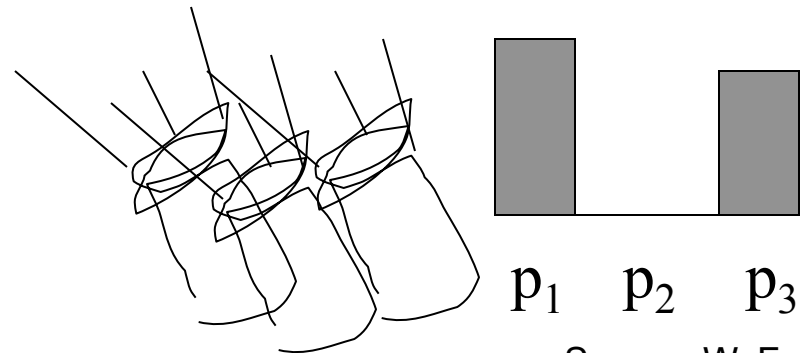
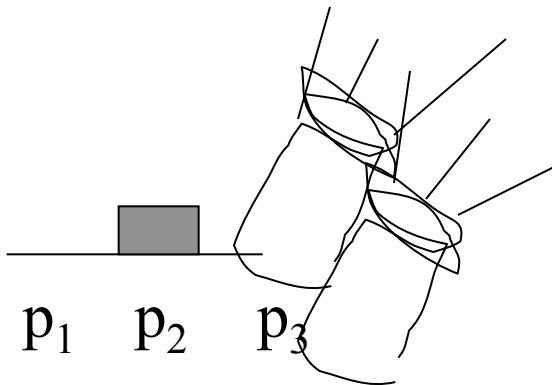
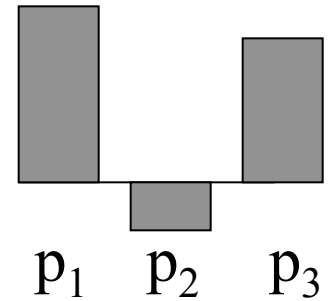
Source: W. Freeman

Color matching experiment 2

We say a “negative” amount of p_2 was needed to make the match, because we added it to the test color’s side.



The primary color amounts needed for a match:



Trichromacy

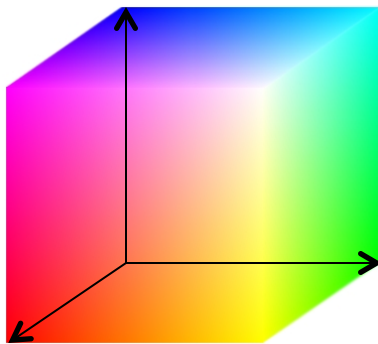
- In color matching experiments, most people can match any given light with three primaries
 - Primaries must be *independent*
- For the same light and same primaries, most people select the same weights
 - Exception: color blindness
- Trichromatic color theory
 - Three numbers seem to be sufficient for encoding color
 - Dates back to 18th century (Thomas Young)

Grassman's Laws

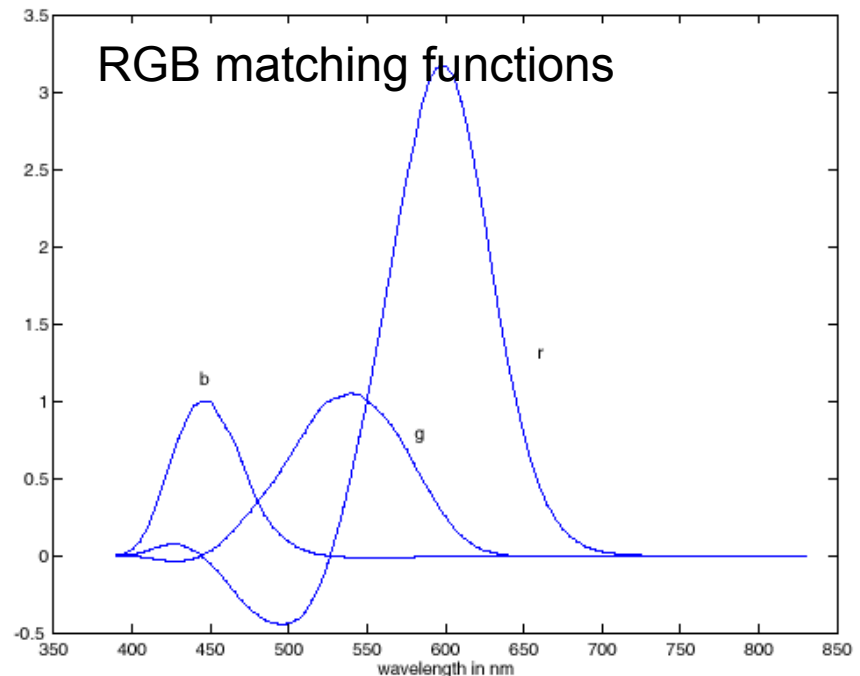
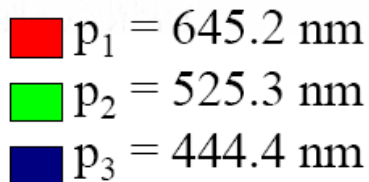
- Color matching appears to be linear
- If two test lights can be matched with the same set of weights, then they match each other:
 - Suppose $A = u_1 P_1 + u_2 P_2 + u_3 P_3$ and $B = u_1 P_1 + u_2 P_2 + u_3 P_3$.
Then $A = B$.
- If we mix two test lights, then mixing the matches will match the result:
 - Suppose $A = u_1 P_1 + u_2 P_2 + u_3 P_3$ and $B = v_1 P_1 + v_2 P_2 + v_3 P_3$.
Then $A + B = (u_1 + v_1) P_1 + (u_2 + v_2) P_2 + (u_3 + v_3) P_3$.
- If we scale the test light, then the matches get scaled by the same amount:
 - Suppose $A = u_1 P_1 + u_2 P_2 + u_3 P_3$.
Then $kA = (ku_1) P_1 + (ku_2) P_2 + (ku_3) P_3$.

Linear color spaces

- Defined by a choice of three *primaries*
- The coordinates of a color are given by the weights of the primaries used to match it
- In addition to primaries, need to specify **matching functions**: the amount of each primary needed to match a monochromatic light source at each wavelength



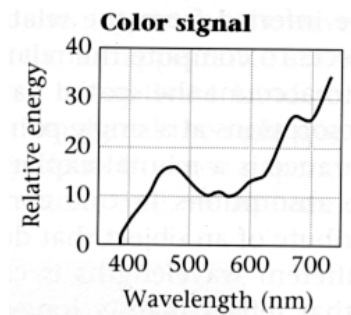
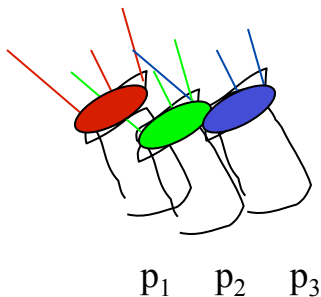
RGB primaries



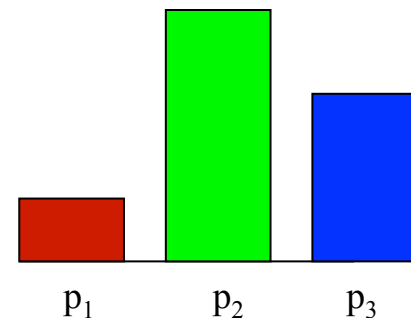
Linear color spaces

- How to compute the weights of the primaries to match any spectral signal?

Given: a choice of three primaries and a target color signal



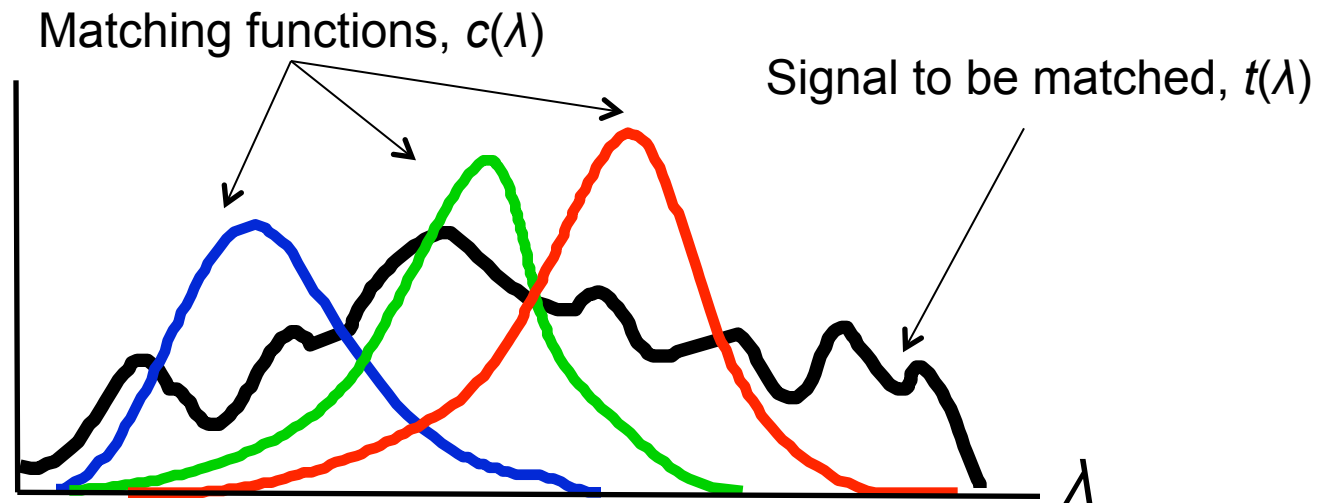
Find: weights of the primaries needed to match the color signal



Linear color spaces

- How to compute the weights of the primaries to match any spectral signal?
- Let $c(\lambda)$ be one of the matching functions, and let $t(\lambda)$ be the spectrum of the signal. Then the weight of the corresponding primary needed to match t is

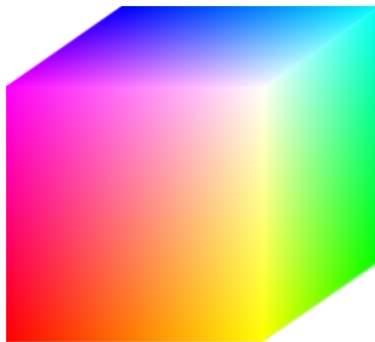
$$w = \int_{\lambda} c(\lambda) t(\lambda) d\lambda$$



RGB space

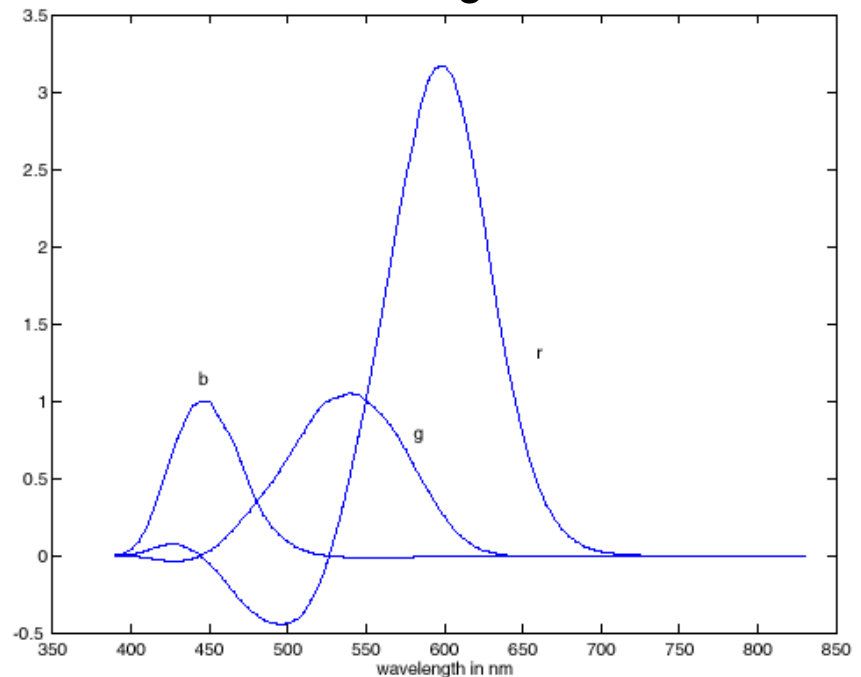
- Primaries are monochromatic lights (for monitors, they correspond to the three types of phosphors)
- *Subtractive matching* required for some wavelengths

RGB primaries



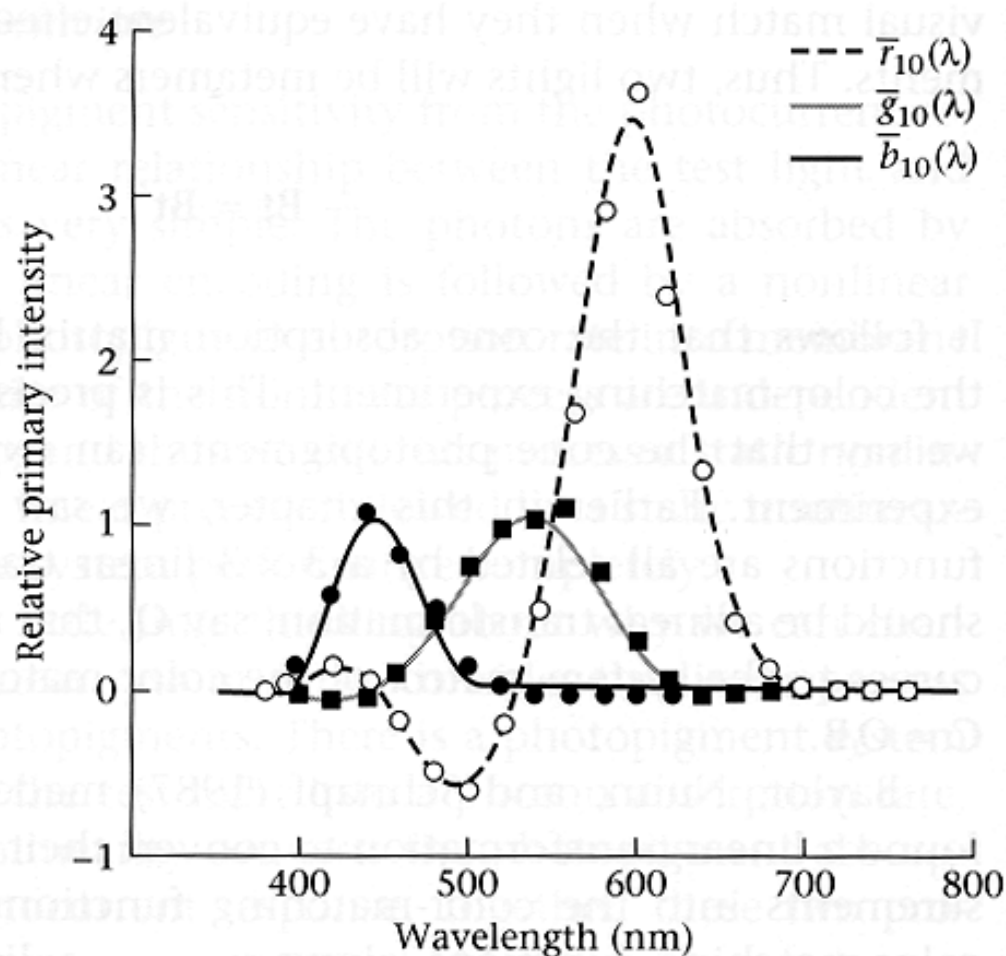
 $p_1 = 645.2 \text{ nm}$
 $p_2 = 525.3 \text{ nm}$
 $p_3 = 444.4 \text{ nm}$

RGB matching functions



Comparison of RGB matching functions with best 3x3 transformation of cone responses

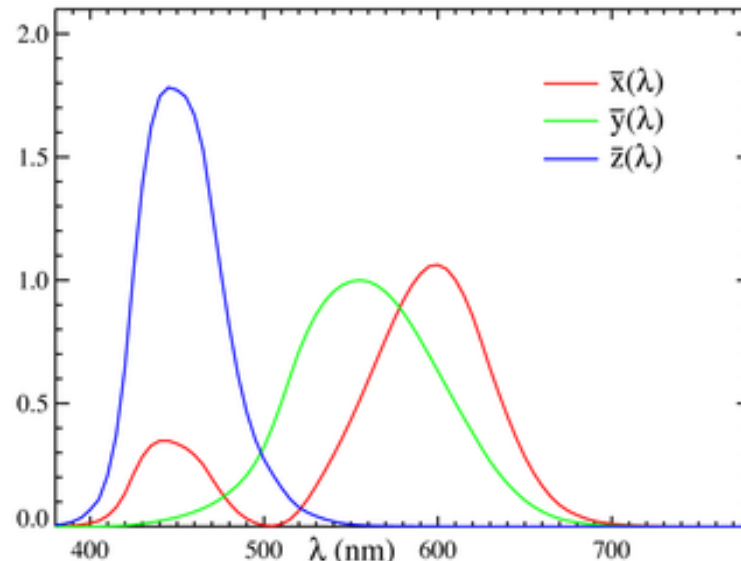
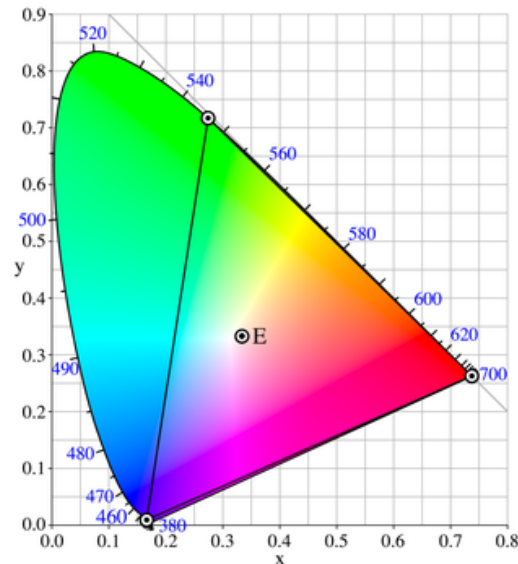
4.20 COMPARISON OF CONE PHOTOCURRENT RESPONSES AND THE COLOR-MATCHING FUNCTIONS. The cone photocurrent spectral responsivities are within a linear transformation of the color-matching functions, after a correction has been made for the optics and inert pigments in the eye. The smooth curves show the Stiles and Burch (1959) color-matching functions. The symbols show the matches predicted from the photocurrents of the three types of macaque cones. The predictions included a correction for absorption by the lens and other inert pigments in the eye. Source: Baylor, 1987.



Linear color spaces: CIE XYZ

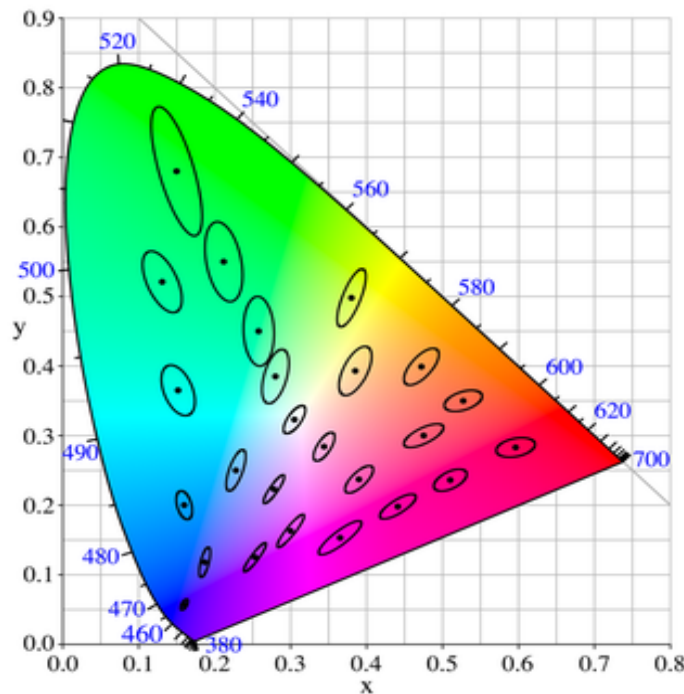
- Primaries are *imaginary*, but matching functions are everywhere positive
- The Y parameter corresponds to brightness or *luminance* of a color
- 2D visualization: draw (x,y) , where $x = X/(X+Y+Z)$, $y = Y/(X+Y+Z)$

Matching functions

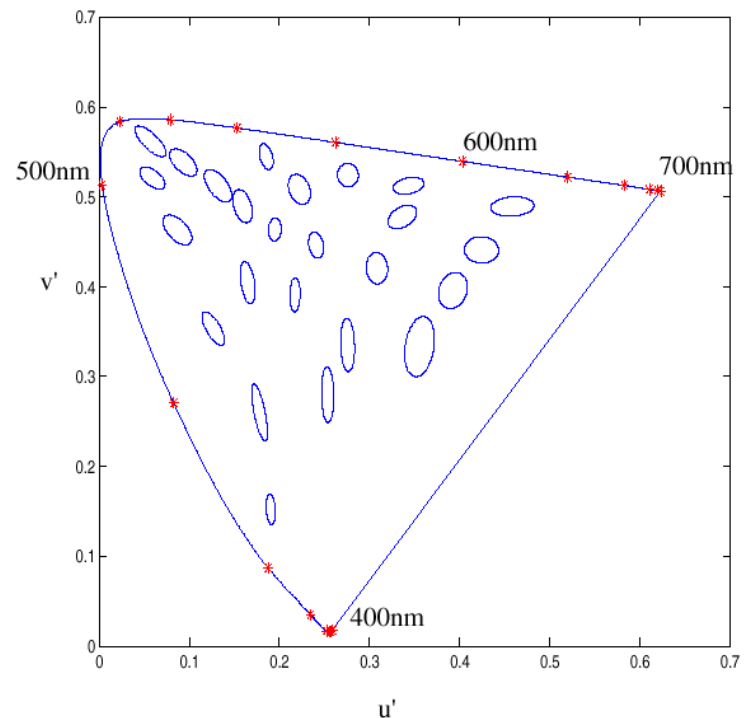


Uniform color spaces

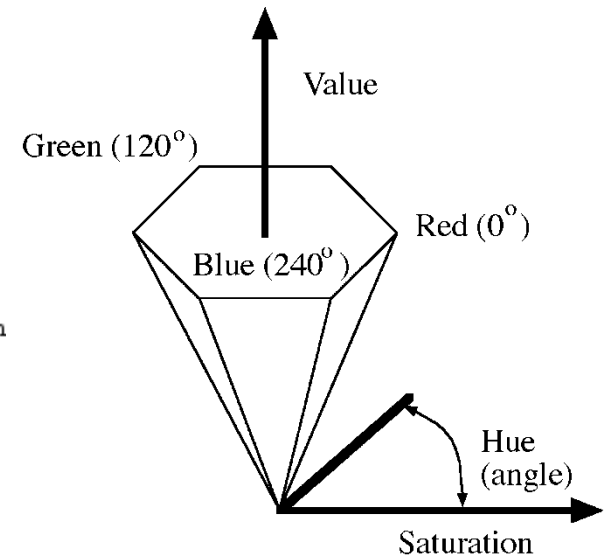
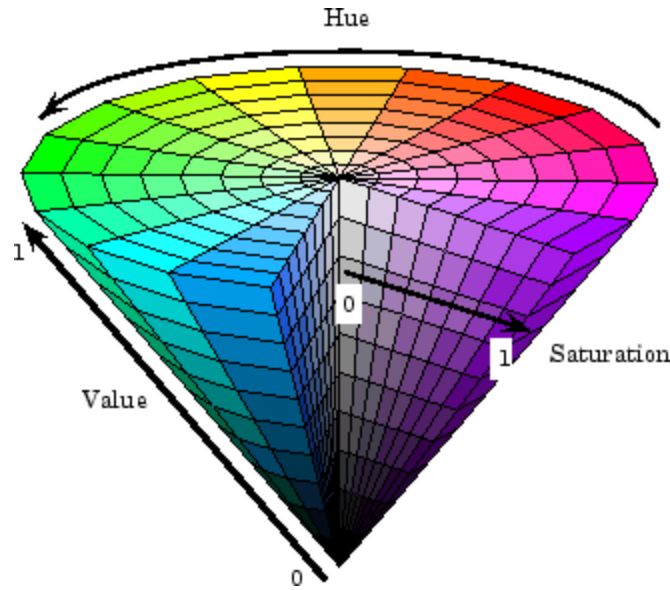
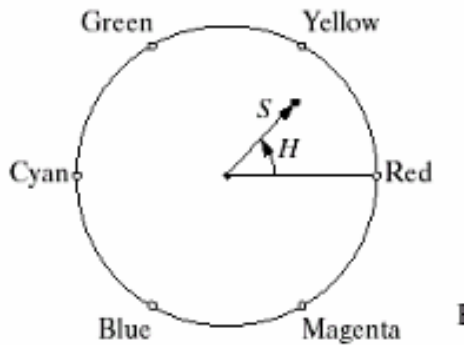
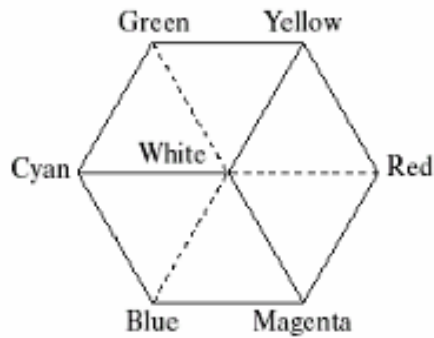
- Unfortunately, differences in x, y coordinates do not reflect perceptual color differences
- CIE $u'v'$ is a projective transform of x, y to make the ellipses more uniform



McAdam ellipses: Just noticeable differences in color



Nonlinear color spaces: HSV



- Perceptually meaningful dimensions: Hue, Saturation, Value (Intensity)
- RGB cube on its vertex

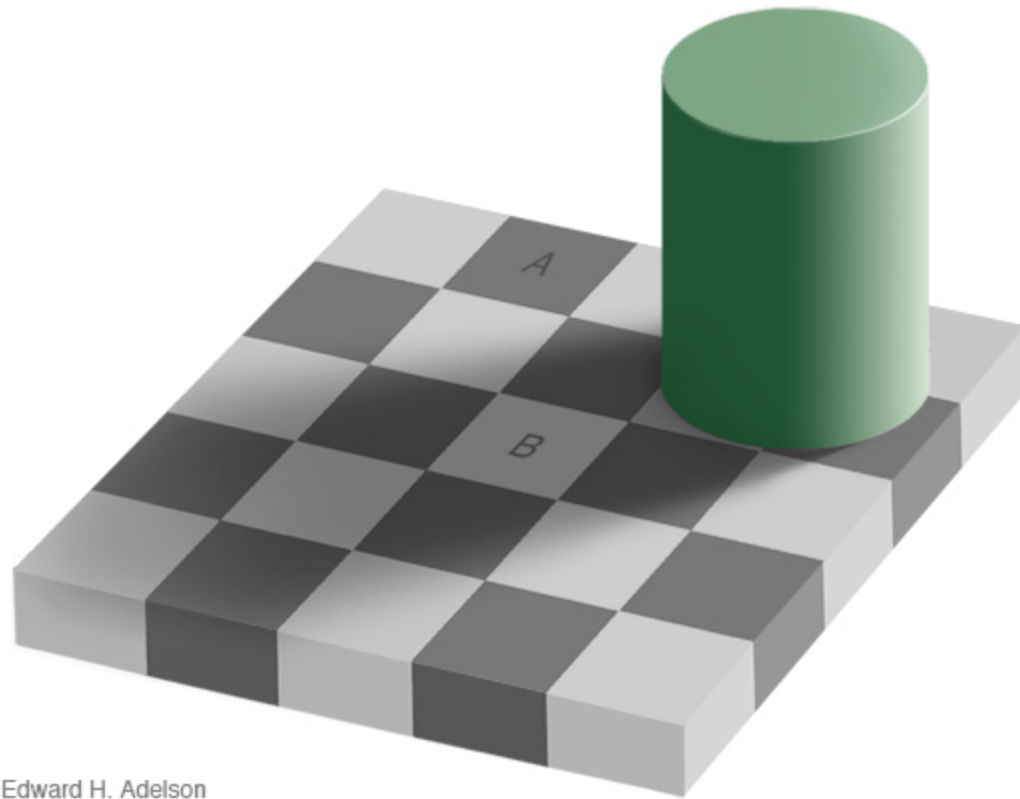
Color perception

- Color/lightness constancy
 - The ability of the human visual system to perceive the intrinsic reflectance properties of the surfaces despite changes in illumination conditions



J. S. Sargent, The Daughters of Edward D. Boit, 1882

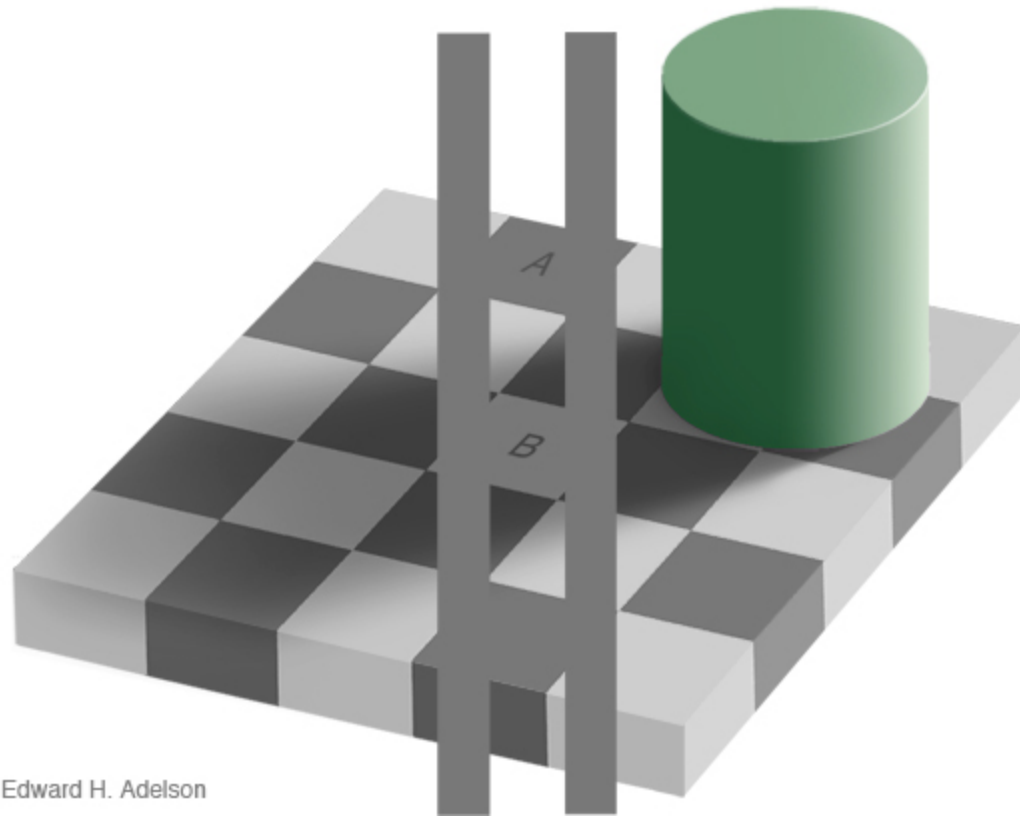
Checker shadow illusion



Edward H. Adelson

http://web.mit.edu/persci/people/adelson/checkershadow_illusion.html

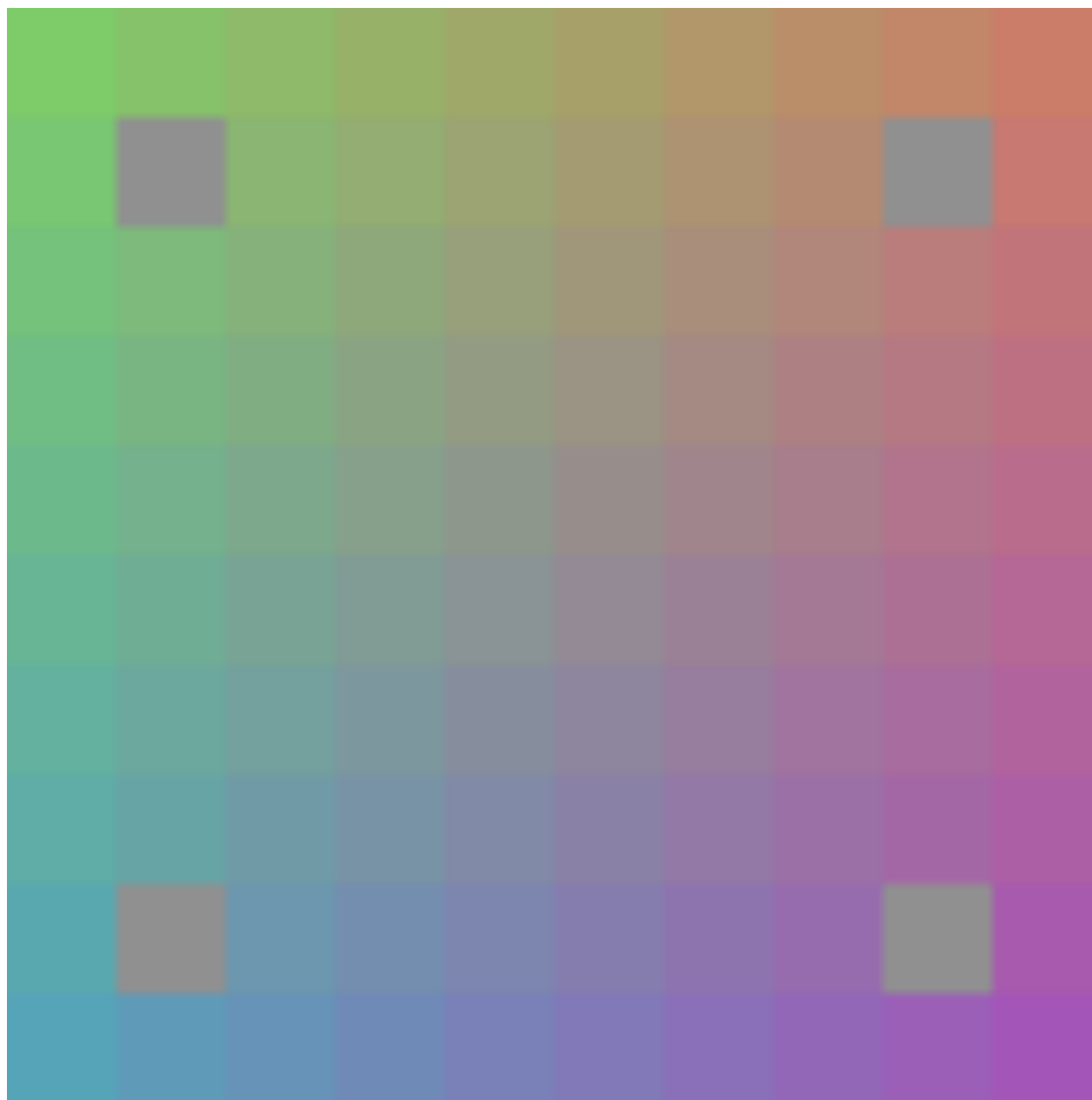
Checker shadow illusion



- Possible explanations
 - Simultaneous contrast
 - Reflectance edges vs. illumination edges

http://web.mit.edu/persci/people/adelson/checkershadow_illusion.html

Simultaneous contrast/Mach bands



Chromatic adaptation

- The visual system changes its sensitivity depending on the luminances prevailing in the visual field
 - The exact mechanism is poorly understood
- Adapting to different brightness levels
 - Changing the size of the iris opening (i.e., the aperture) changes the amount of light that can enter the eye
 - Think of walking into a building from full sunshine
- Adapting to different color temperature
 - The receptive cells on the retina change their sensitivity
 - For example: if there is an increased amount of red light, the cells receptive to red decrease their sensitivity until the scene looks white again
 - We actually adapt better in brighter scenes: This is why candlelit scenes still look yellow

White balance

- When looking at a picture on screen or print, our eyes are adapted to the illuminant of the room, not to that of the scene in the picture
- When the white balance is not correct, the picture will have an unnatural color “cast”

incorrect white balance



correct white balance



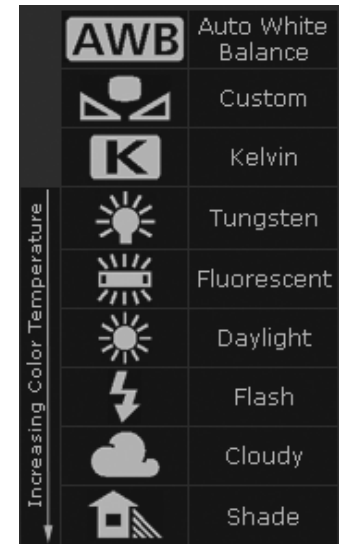
White balance



<http://www.wired.com/2015/02/science-one-agrees-color-dress/>

White balance

- Film cameras:
 - Different types of film or different filters for different illumination conditions
- Digital cameras:
 - Automatic white balance
 - White balance settings corresponding to several common illuminants
 - Custom white balance using a reference object



White balance

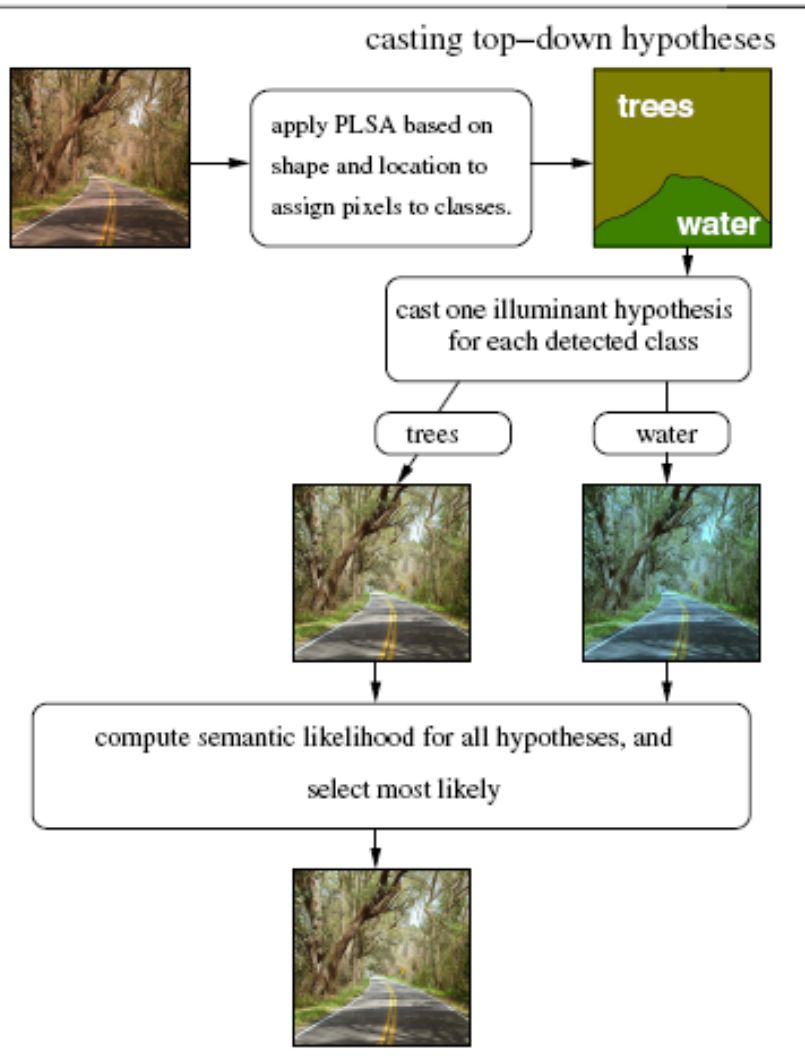
- Von Kries adaptation
 - Multiply each channel by a gain factor
- Best way: gray card
 - Take a picture of a neutral object (white or gray)
 - Deduce the weight of each channel
 - If the object is recoded as r_w , g_w , b_w
use weights $1/r_w$, $1/g_w$, $1/b_w$



White balance

- Without gray cards: we need to “guess” which pixels correspond to white objects
- Gray world assumption
 - The image average r_{ave} , g_{ave} , b_{ave} is gray
 - Use weights $1/r_{ave}$, $1/g_{ave}$, $1/b_{ave}$
- Brightest pixel assumption
 - Highlights usually have the color of the light source
 - Use weights inversely proportional to the values of the brightest pixels
- Gamut mapping
 - Gamut: convex hull of all pixel colors in an image
 - Find the transformation that matches the gamut of the image to the gamut of a “typical” image under white light
- Use image statistics, learning techniques

White balance by recognition



- Key idea: For each of the semantic classes present in the image, compute the illuminant that transforms the pixels assigned to that class so that the average color of that class matches the average color of the same class in a database of “typical” images

Mixed illumination

- When there are several types of illuminants in the scene, different reference points will yield different results



Reference: moon



Reference: stone

Spatially varying white balance



Input



Alpha map



Output

E. Hsu, T. Mertens, S. Paris, S. Avidan, and F. Durand,
[Light Mixture Estimation for Spatially Varying White Balance](#),
SIGGRAPH 2008

Color cues in computer vision

Image description
and matching



Image segmentation

